Tax Multipliers in a DSGE Model*

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Abstract

This paper estimates fiscal multipliers for a broad set of tax instruments in a medium-scale DSGE model. The model includes labor, capital, consumption, and dividend taxes, an investment tax credit, and expensing of investment. We estimate the model with Bayesian methods on U.S. data from 1961 to 2013. Instruments that lower the cost of capital have the largest multipliers: per dollar of forgone revenue, a cut in the capital income tax raises the present value of output by about 2.8 dollars over three years, and the investment tax credit by about 2.2 dollars. Cuts to labor and consumption taxes have smaller multipliers, and the consumption tax multiplier is small at every horizon. When the resulting debt is repaid partly through distortionary taxes, several multipliers turn negative within five years; under fiscal rules in which only transfers respond to debt, they stay positive; highlighting the importance of fiscal feedback in determining the effectiveness of tax policies. Allowing the price of capital goods to rise with investment lowers the investment tax credit multiplier by about a quarter. Estimates for the United Kingdom, under fiscal rules in which only transfers respond to debt, are broadly similar to the corresponding U.S. estimates.

1 Introduction

We study tax multipliers including those associated with taxes that directly stimulate private consumption spending or investment such as consumption taxes and investment tax credits. Consumption taxes have long supported by economists for their relative efficiency vis-a-vis taxes on capital and labor ((Correia, 2010), (da Costa and Santos, 2023)). This paper looks at the stabilization properties on consumption taxes, as well as other tax instruments, such as investment tax credits and changes to expensing and depreciation policies. These policies target investment by lowering the after-tax cost of capital. Historically, such instruments have been used by the U.S. to provide counter-cyclical fiscal policy (e.g. the Economic Recovery Act of 1981 increased the ability of businesses to take advantage of the investment tax credit). Despite their use, such policies have not been studied in a DSGE environment.

A number of researchers have used DSGE models to study the government spending multiplier (e.g., Christiano et al. (2010)), but less work has been done on tax multipliers. Those who provide insights on tax multipliers (e.g., Zubairy (2014), Kumhof et al. (2010)) focus on multipliers associated with broad taxes on capital and labor income.

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In order to compare to the past literature on tax multipliers, we use a standard DSGE framework and solution method. Our only changes to the standard approach are to include additional tax instruments: consumption taxes, the investment tax credit, and capital cost recovery rules. In our baseline model every fiscal instrument follows a rule that responds to lagged debt and output.

Three findings from our study stand out. First, the instruments that act directly on the cost of capital, the capital income tax and the investment tax credit, have the largest multipliers at every horizon beyond the first year. Second, the consumption tax multiplier is small, which seems to reflect that in a representative-agent model the consumption tax works only through the consumption-leisure margin. Third, how the increase in debt associated with a tax cut is financed matters a great deal at longer horizons: with debt repaid partly through distortionary taxes, the spending, consumption tax, and expensing multipliers turn negative after three to four years, while under fiscal rules in which only lump-sum transfers respond to debt they remain positive. We also show that a capital-goods supply friction in the spirit of House and Shapiro (2008) mainly lowers the multipliers of the investment incentives, and we report estimates for the United Kingdom.

The remainder of the paper is organized as follows. Section 2 describes the model. Section 2.8 describes the calibration and estimation strategy we use to identify parameter values. Section 3 presents the results. Section 4 compares alternative fiscal rules and the capital-goods friction, and Section 7 reports the sensitivity to calibrated parameters. Section 8 concludes. The appendix reports the estimates under alternative specifications and for the United Kingdom.

2 Model

2.1 Households

The economy is populated by a continuum of identical households. Each household maximizes expected lifetime utility over its habit-adjusted private consumption surplus $x_{c,t}$, hours worked h_t, and habit-adjusted government consumption surplus $x_{g,t}$:

$$E_0 \sum_{t=0}^{\infty} \beta^t d_t \left\{ \frac{\left[x_{c,t}^a (1-h_t)^{1-a}\right]^{1-\gamma}}{1-\gamma} + V(x_{g,t}) \right\} \tag{2.1}$$

where $\beta \in (0,1)$ is the discount factor, d_t is an intertemporal preference shock, γ is the coefficient of relative risk aversion, and $a \in (0,1)$ is the consumption expenditure share calibrated at the steady state. Utility over government consumption $V(x_{g,t})$ is separable from private utility, so government spending does not directly affect the marginal utility of private consumption or leisure. Following Zubairy (2014), the model incorporates deep habit formation in both private and government consumption as in Ravn et al. (2006), which generates counter-cyclical markups and allows private consumption to rise in response to a government spending shock.

The habit-adjusted consumption surpluses are:

$$x_{c,t} = c_t - b_c s_{C,t-1} \tag{2.2}$$

$$x_{g,t} = g_t - b_g s_{G,t-1} \tag{2.3}$$

where $b_c \in [0,1)$ and $b_g \in [0,1)$ govern the degree of deep habit formation in private and government consumption. The habit stocks evolve according to:

$$s_{C,t} = \theta^c s_{C,t-1} + (1-\theta^c)c_t \tag{2.4}$$

$$s_{G,t} = \theta^g s_{G,t-1} + (1-\theta^g)g_t \tag{2.5}$$

where $\theta^c, \theta^g \in [0,1)$ govern the speed of habit stock adjustment.

Households supply labor, accumulate physical capital, hold government bonds, and receive dividend income from firms. The period-by-period real budget constraint is:

$$\begin{aligned}(1+\tau_t^c)c_t + b_t + i_t = \frac{\left[1+(R_{t-1}-1)(1-\tau_t^i)\right]b_{t-1}}{\pi_t} + (1-\tau_t^w)w_t h_t \\ + (1-\tau_t^k)r_t^k u_t k_{t-1} + \tau_t^k \delta^\tau k_{t-1}^\tau + \tau_t^k e_t^\tau i_t + \tau_t^{ic} i_t + (1-\tau_t^d)\,\mathrm{div}_t + x_t\end{aligned} \tag{2.6}$$

where b_t denotes real bond holdings, R_t is the gross nominal interest rate, $\pi_t \equiv P_t/P_{t-1}$ is gross price inflation, w_t is the real wage, r_t^k is the real rental rate of capital, u_t is the capital utilization rate, div_t are real firm dividends, and x_t is a lump-sum government transfer. The tax instruments are: τ_t^c (consumption tax), τ_t^w (labor income tax), τ_t^k (capital income tax), τ_t^i (interest income tax), τ_t^d (dividend income tax), and τ_t^{ic} (investment tax credit). Only the interest component $(R_{t-1}-1)$ of the gross bond return is subject to the interest income tax. Capital income is measured net of tax depreciation deductions $\tau_t^k \delta^\tau k_{t-1}^\tau$, where δ^τ is the tax depreciation rate and k_t^τ is the tax basis of the capital stock, and net of immediate expensing $\tau_t^k e_t^\tau i_t$, where e_t^τ is the expensing rate.

Physical capital accumulates according to:

$$k_t = (1-\delta)k_{t-1} + i_t\left[1 - S\left(\frac{\mu_t i_t}{i_{t-1}}\right)\right] \tag{2.7}$$

where δ is the constant economic depreciation rate and μ_t is a stochastic investment efficiency shock. Following Smets and Wouters (2003) and Zubairy (2014), the investment adjustment cost function takes the form $S\left(\frac{\mu_t i_t}{i_{t-1}}\right) = \frac{\kappa}{2}\left(\frac{\mu_t i_t}{i_{t-1}} - 1\right)^2$, so that in the deterministic steady state $S = S' = 0$ and $S'' = \kappa > 0$.

The tax basis of the capital stock evolves as:

$$k_t^\tau = (1-\delta^\tau)k_{t-1}^\tau + i_t(1-e_t^\tau) \tag{2.8}$$

Only the non-expensed fraction $(1-e_t^\tau)$ of new investment is added to the tax basis; the immediately expensed portion is deducted in the period of investment.

Household optimality conditions

The first-order condition with respect to consumption defines the marginal utility of income λ_t:

$$\lambda_t = \frac{d_t \cdot a \cdot \left[x_{c,t}^a (1-h_t)^{1-a}\right]^{1-\gamma}}{x_{c,t}(1+\tau_t^c)} \tag{2.9}$$

The consumption tax τ_t^c drives a wedge between the marginal utility of income and the marginal utility of the consumption surplus: a higher τ_t^c lowers λ_t for any given level of $x_{c,t}$ and h_t.

The intratemporal labor-leisure condition equates the marginal disutility of work to the after-markup real wage:

$$d_t(1-a)\frac{\left[x_{c,t}^{a}(1-h_t)^{1-a}\right]^{1-\gamma}}{1-h_t} = \frac{\lambda_t w_t}{\tilde{w}_t} \tag{2.10}$$

where $\tilde{w}_t$ is the wage markup arising from monopolistic competition in the labor market. The labor income tax τ_t^w does not appear explicitly here because it is absorbed into the endogenous markup: the wage Phillips curve (2.28) implies $\tilde{w}_t = \eta_w/[(\eta_w - 1)(1-\tau_t^w)]$ in the frictionless limit, so the effective wage $w_t/\tilde{w}_t$ already reflects the after-tax, after-markup return to labor.

The optimal utilization rate equates the after-tax marginal product of capital to the marginal cost of utilization:

$$(1-\tau_t^k)r_t^k = \gamma_1 u_t^{\sigma_u} \tag{2.11}$$

where $\gamma_1 = (1-\tau_k^{ss})r_k^{ss}$ is pinned down at the steady state and σ_u governs the convexity of the utilization cost function $a(u_t) = \frac{\gamma_1}{1+\sigma_u}(u_t^{1+\sigma_u} - 1)$. Relative to Zubairy (2014), the term $\delta q_t \tau_t^k$ is absent from this condition because depreciation deductions are now tracked through the shadow value of the tax basis $v_{\tau,t}$ defined below, avoiding double-counting.

The capital Euler equation governs the optimal capital stock:

$$\lambda_t q_t = \beta E_t \lambda_{t+1}\left[(1-\tau_{t+1}^k)r_{t+1}^k u_{t+1} - \frac{\gamma_1}{1+\sigma_u}\left(u_{t+1}^{1+\sigma_u} - 1\right) + q_{t+1}(1-\delta)\right] \tag{2.12}$$

where q_t is Tobin's marginal q. Similarly, the term $\delta q_{t+1}\tau_{t+1}^k u_{t+1}$ present in Zubairy (2014) is omitted here for the same reason.

The investment Euler equation governs the optimal investment decision:

$$\lambda_t\left(1-\tau_t^{ic}-\tau_t^k e_t^\tau\right) = \lambda_t q_t\left[1 - S\left(\tfrac{\mu_t i_t}{i_{t-1}}\right) - \tfrac{\mu_t i_t}{i_{t-1}}S'\left(\tfrac{\mu_t i_t}{i_{t-1}}\right)\right] + \beta E_t \lambda_{t+1} q_{t+1}\left(\tfrac{\mu_{t+1} i_{t+1}}{i_t}\right)^2 S'\left(\tfrac{\mu_{t+1} i_{t+1}}{i_t}\right) + \lambda_t v_{\tau,t}(1-e_t^\tau) \tag{2.13}$$

The left-hand side is the net cost of investing one unit: the investment tax credit τ_t^{ic} and the immediate expensing saving $\tau_t^k e_t^\tau$ each reduce the effective outlay. The term $\lambda_t v_{\tau,t}(1-e_t^\tau)$ on the right-hand side is the present value of future tax depreciation deductions on the non-expensed share of investment. The shadow value of the tax basis $v_{\tau,t}$ satisfies its own Euler equation:

$$\lambda_t v_{\tau,t} = \beta E_t \lambda_{t+1}\left[(1-\delta^\tau)v_{\tau,t+1} + \tau_{t+1}^k \delta^\tau\right] \tag{2.14}$$

An additional unit of tax basis yields a deduction $\tau_{t+1}^k \delta^\tau$ next period and a depreciated stock $(1-\delta^\tau)v_{\tau,t+1}$ of future deductions.

Finally, the bond Euler equation is:

$$\lambda_t = \beta E_t\left[1 + (R_t - 1)(1-\tau_{t+1}^i)\right]\frac{\lambda_{t+1}}{\pi_{t+1}} \tag{2.15}$$

Only the interest earned $(R_t - 1)$ on bond holdings is subject to the interest income tax τ_{t+1}^i, so the relevant return is the after-tax net interest. This differs from Zubairy (2014), in which the pre-tax gross return R_t governs the bond Euler.

2.2 Firms

2.2.1 Final Goods Producers

A continuum of monopolistically competitive firms indexed by $i \in [0,1]$ each produce a differentiated variety of output demanded by households, the government, and for investment. Because households and the government form deep habits over individual varieties (Section 2), the demand for variety i in the private consumption and government markets is:

$$c_{i,t} = \left(\frac{P_{i,t}}{P_t}\right)^{-\eta} x_{c,t} + b_c s_{C,i,t-1} \tag{2.16}$$

$$g_{i,t} = \left(\frac{P_{i,t}}{P_t}\right)^{-\eta} x_{g,t} + b_g s_{G,i,t-1} \tag{2.17}$$

Investment demand for variety i is $i_{i,t} = (P_{i,t}/P_t)^{-\eta}(y_t - c_t - g_t)$; there are no deep habits in investment. Here $\eta > 1$ is the elasticity of substitution across varieties and P_t is the aggregate price index. The habit-inelastic components $b_c s_{C,i,t-1}$ and $b_g s_{G,i,t-1}$ make a portion of demand perfectly price-inelastic, generating counter-cyclical markups: when aggregate demand expands, the price-elastic share rises, increasing effective demand elasticity and compressing markups. Firms therefore invest in their customer base by holding prices low today to earn higher profits on a larger habit stock in the future.

2.2.2 Intermediate Goods Producers

Each firm i produces output using capital services $u_t k_{i,t-1}$ and labor $h_{i,t}$ according to a Cobb-Douglas technology with a fixed operating cost ψ:

$$\tilde{y}_{i,t} = z_t (u_t k_{i,t-1})^{\theta} h_{i,t}^{1-\theta} - \psi \tag{2.18}$$

where z_t is an aggregate technology shock, θ is the capital income share, and ψ is calibrated to match a realistic profit-to-output ratio in steady state. The technology shock follows an AR(1) process:

$$\log z_t = \rho_z \log z_{t-1} + \varepsilon_t^z, \quad \varepsilon_t^z \sim \mathcal{N}(0, \sigma_z^2) \tag{2.19}$$

where $\rho_z \in [0,1)$. Cost minimization at given factor prices yields the symmetric-equilibrium factor demand conditions:

$$w_t = mc_t \cdot z_t (1-\theta)(u_t k_{t-1})^{\theta} h_t^{-\theta} \tag{2.20}$$

$$r_t^k = mc_t \cdot z_t \theta (u_t k_{t-1})^{\theta-1} h_t^{1-\theta} \tag{2.21}$$

where mc_t is the common real marginal cost across firms.

Following Rotemberg (1982), each firm faces a quadratic cost of changing its nominal price, which reduces output available for final demand. In symmetric equilibrium, aggregate output net of price adjustment costs is:

$$y_t = z_t (u_t k_{t-1})^{\theta} h_t^{1-\theta} - \psi - \frac{\alpha_P}{2}(\pi_t - \bar{\pi})^2 \tag{2.22}$$

where $\alpha_P > 0$ governs the magnitude of price adjustment costs and $\bar{\pi}$ is the steady-state gross inflation rate.

Because deep habits create dynamic pricing incentives, each firm tracks the shadow value of its customer base in each demand segment. In symmetric equilibrium, the shadow values $\tilde{\nu}_t^c$, $\tilde{\nu}_t^g$, and $\tilde{\nu}_t^i$ for the private consumption, government consumption, and investment markets satisfy:

$$1 - mc_t - \tilde{\nu}_t^i = 0 \tag{2.23}$$

$$\frac{1 - mc_t - \tilde{\nu}_t^c}{\theta^c - 1} = \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \left[b_c \tilde{\nu}_{t+1}^c + \frac{\theta^c}{\theta^c - 1} \left(1 - mc_{t+1} - \tilde{\nu}_{t+1}^c\right) \right] \tag{2.24}$$

$$\frac{1 - mc_t - \tilde{\nu}_t^g}{\theta^g - 1} = \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \left[b_g \tilde{\nu}_{t+1}^g + \frac{\theta^g}{\theta^g - 1} \left(1 - mc_{t+1} - \tilde{\nu}_{t+1}^g\right) \right] \tag{2.25}$$

Equation (2.23) states that the investment market, lacking habit-inelastic demand, yields a zero effective markup. Equations (2.24)–(2.25) are forward-looking Euler equations: a firm accepts a lower current markup to build its habit stock, earning higher future rents.

These conditions together with the demand structure yield the aggregate price Phillips curve:

$$\eta\big(\tilde{\nu}_t^c x_{c,t} + \tilde{\nu}_t^g x_{g,t} + \tilde{\nu}_t^i (y_t - c_t - g_t)\big) + \alpha_P \pi_t (\pi_t - \bar{\pi}) - y_t = \alpha_P \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \pi_{t+1} (\pi_{t+1} - \bar{\pi}) \tag{2.26}$$

Each household is also a monopolistic supplier of a differentiated labor variety and faces quadratic wage adjustment costs. Defining gross wage inflation $\pi_t^w \equiv (w_t / w_{t-1}) \pi_t$, the wage Phillips curve is:

$$\pi_t^w = \frac{w_t}{w_{t-1}} \pi_t \tag{2.27}$$

$$(\eta_w - 1)(1 - \tau_t^w) h_t + \alpha_W \pi_t^w (\pi_t^w - \bar{\pi}) - \frac{\eta_w h_t}{\tilde{w}_t} = \alpha_W \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \pi_{t+1}^w (\pi_{t+1}^w - \bar{\pi}) \tag{2.28}$$

where $\eta_w > 1$ is the elasticity of substitution across labor varieties and α_W governs wage adjustment costs. The labor income tax τ_t^w enters the wage Phillips curve because it affects the effective value of the marginal product of labor to the household.

Firm dividends are the residual of net output after paying all factors:

$$\text{div}_t = y_t - w_t h_t - r_t^k u_t k_{t-1} \tag{2.29}$$

The aggregate resource constraint equates gross production (net of fixed costs) to all expenditure uses plus utilization and price and wage adjustment costs:

$$z_t (u_t k_{t-1})^\theta h_t^{1-\theta} - \psi = c_t + g_t + i_t + \frac{\gamma_1}{1 + \sigma_u} (u_t^{1+\sigma_u} - 1) k_{t-1} + \frac{\alpha_P}{2} (\pi_t - \bar{\pi})^2 + \frac{\alpha_W}{2} (\pi_t^w - \bar{\pi})^2 w_t \tag{2.30}$$

2.3 Government/Fiscal Authority

The government issues one-period nominal bonds, purchases goods, pays lump-sum transfers to households, and collects tax revenue from all six tax instruments. The real government budget constraint is:

$$b_t = \frac{R_{t-1}\, b_{t-1}}{\pi_t} + g_t + x_t - taxrev_t + \varepsilon_t^b \tag{2.31}$$

where b_t is real government debt, g_t is real government purchases, x_t is real lump-sum transfers, $taxrev_t$ is total real tax revenue, and ε_t^b is an i.i.d. residual added to the budget identity. Tax revenue is:

$$\begin{aligned} taxrev_t = {} & \tau_t^c c_t + \tau_t^w w_t h_t + \tau_t^k \Big(r_t^k u_t k_{t-1} - \delta^\tau k_{t-1}^\tau - e_t^\tau i_t \Big) \\ & + \tau_t^i (R_{t-1} - 1) \frac{b_{t-1}}{\pi_t} + \tau_t^d \, \mathrm{div}_t - \tau_t^{ic} \, i_t \end{aligned} \tag{2.32}$$

The six components are: consumption tax revenue ($\tau_t^c c_t$), labor income tax revenue ($\tau_t^w w_t h_t$), capital income tax revenue net of tax depreciation deductions and immediate expensing ($\tau_t^k(\cdot)$), interest income tax revenue on bond interest ($\tau_t^i(R_{t-1}-1)b_{t-1}/\pi_t$), dividend tax revenue ($\tau_t^d \, \mathrm{div}_t$), and investment tax credit outlays ($-\tau_t^{ic} \, i_t$).

Fiscal rules

Each fiscal instrument follows an autoregressive rule that allows feedback from lagged government debt and lagged output, following Zubairy (2014). Debt feedback ensures intertemporal fiscal solvency; output feedback captures automatic stabilizers. Because g_t and x_t are expenditures that the government cuts when debt is elevated, their debt-feedback coefficients enter with a negative sign. Tax rates carry a positive debt-feedback sign, rising to close the fiscal gap.

Government purchases and lump-sum transfers follow:

$$\log\left(\frac{g_t}{\bar{g}}\right) = \rho_g \log\left(\frac{g_{t-1}}{\bar{g}}\right) - \rho_{g,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{g,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^g \tag{2.33}$$

$$\log\left(\frac{x_t}{\bar{x}}\right) = \rho_x \log\left(\frac{x_{t-1}}{\bar{x}}\right) - \rho_{x,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{x,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^x \tag{2.34}$$

The rules for the capital and labor income tax rates, which follow Zubairy (2014), are:

$$\log\left(\frac{\tau_t^k}{\bar{\tau}^k}\right) = \rho_k \log\left(\frac{\tau_{t-1}^k}{\bar{\tau}^k}\right) + \rho_{k,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{k,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^k \tag{2.35}$$

$$\log\left(\frac{\tau_t^w}{\bar{\tau}^w}\right) = \rho_w \log\left(\frac{\tau_{t-1}^w}{\bar{\tau}^w}\right) + \rho_{w,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{w,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \phi \, \varepsilon_t^k + \varepsilon_t^w \tag{2.36}$$

The term $\phi \, \varepsilon_t^k$ in the labor tax rule allows capital tax shocks to generate correlated movements in the labor tax rate, capturing the empirical co-movement between capital and labor income tax changes.

The consumption and dividend tax rules take the same log-ratio form:

$$\log\left(\frac{\tau_t^c}{\bar{\tau}^c}\right) = \rho_c \log\left(\frac{\tau_{t-1}^c}{\bar{\tau}^c}\right) + \rho_{c,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{c,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^c \tag{2.37}$$

$$\log\left(\frac{\tau_t^d}{\bar{\tau}^d}\right) = \rho_d^\tau \log\left(\frac{\tau_{t-1}^d}{\bar{\tau}^d}\right) + \rho_{d,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{d,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^d \tag{2.38}$$

The rules for the investment tax credit and the expensing rate are specified in level-deviation form because the log-ratio form is numerically unstable when the underlying rate can take

values near zero:

$$\tau_t^{ic} - \bar{\tau}^{ic} = \rho_{ic}(\tau_{t-1}^{ic} - \bar{\tau}^{ic}) + \rho_{ic,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{ic,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^{ic} \tag{2.39}$$

$$e_t^{\tau} - \bar{e}^{\tau} = \rho_e(e_{t-1}^{\tau} - \bar{e}^{\tau}) + \rho_{e,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \rho_{e,y} \log\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_t^{e} \tag{2.40}$$

The interest income tax is held constant at its steady-state value, $\tau_t^i = \bar{\tau}^i$. The interest income tax rate is not among the observables (Section 2.8.2), so the parameters of a rule for it would be identified only through cross-equation restrictions, and in practice they appear to be poorly identified. We therefore keep the rate in the budget constraint and the bond Euler equation but do not let it move.

AR(1) fiscal rules

As an alternative to the rules above, we also estimate the model with AR(1) fiscal rules in which lump-sum transfers are the only instrument that responds to debt. In this version each tax rate, government purchases, the investment tax credit, and the expensing rate follow equations (2.33)–(2.40) with all debt and output feedback coefficients set to zero, and the transfer rule (2.34) keeps its debt feedback but drops its output feedback:

$$\log\left(\frac{x_t}{\bar{x}}\right) = \rho_x \log\left(\frac{x_{t-1}}{\bar{x}}\right) - \rho_{x,b} \log\left(\frac{b_{t-1}}{\bar{b}}\right) + \varepsilon_t^x \tag{2.41}$$

The debt created by any shock is then repaid through lump-sum transfers alone, so the comparison with the baseline rules isolates the role of distortionary tax financing (Section 4).

2.4 Monetary Authority

The monetary authority sets the gross nominal interest rate R_t according to a Taylor rule with interest rate smoothing, following Zubairy (2014):

$$\log\left(\frac{R_t}{\bar{R}}\right) = \alpha_R \log\left(\frac{R_{t-1}}{\bar{R}}\right) + (1-\alpha_R)\left[\alpha_\pi \log\left(\frac{\pi_t}{\bar{\pi}}\right) + \alpha_Y \log\left(\frac{y_t}{\bar{y}}\right)\right] + \varepsilon_t^m \tag{2.42}$$

where $\alpha_R \in (0,1)$ governs interest rate inertia, $\alpha_\pi > 1$ is the response to inflation deviations from steady state (satisfying the Taylor principle), $\alpha_Y > 0$ is the response to output deviations, and ε_t^m is a monetary policy shock. The interest income tax τ_t^i does not enter the Taylor rule directly; it enters the household bond Euler equation (2.15), which links the pre-tax policy rate R_t set by the central bank to the after-tax return relevant for household saving decisions. Since τ_t^i is held constant in the estimated models, it affects this link only through the steady state.

2.5 Extension: Capital-Goods Supply Friction

In the baseline model one unit of output can always be turned into one unit of investment, so the relative price of investment goods is fixed at one. House and Shapiro (2008) find that the supply of capital goods is not perfectly elastic in the short run, so that part of a temporary investment incentive is absorbed by a higher price of capital goods rather than a higher quantity. To capture this, we add a competitive capital-goods sector that turns

final output into investment goods at a convex cost. Producing i_t units of investment goods uses

$$\Xi_t = \bar{\Xi}\left(\frac{i_t}{\bar{\imath}}\right)^{1+1/\epsilon_K} + F_K \tag{2.43}$$

units of output, where $\epsilon_K > 0$ is the supply elasticity of capital goods and F_K is a fixed cost. Profit maximization at the relative price P_t^I gives the inverse supply curve

$$P_t^I = \left(\frac{i_t}{\bar{\imath}}\right)^{1/\epsilon_K} \tag{2.44}$$

The constants $\bar{\Xi}$ and F_K are set so that the sector earns zero profit in steady state and absorbs exactly $\bar{\imath}$ units of output there, so the steady state is identical to the baseline model and the friction matters only away from it. Households now pay P_t^I per unit of investment: the left-hand side of the investment Euler equation (2.13) becomes $\lambda_t P_t^I(1-\tau_t^{ic}-\tau_t^k e_t^\tau)$, the value of future depreciation deductions is $\lambda_t v_{\tau,t}(1-e_t^\tau)P_t^I$, and the tax basis accumulates the purchase value, $k_t^\tau = (1-\delta^\tau)k_{t-1}^\tau + P_t^I i_t(1-e_t^\tau)$. The investment tax credit and expensing apply to the purchase value $P_t^I i_t$, Ξ_t replaces i_t in the resource constraint, and the sector's profits $P_t^I i_t - \Xi_t$ are paid to households as dividends. As $\epsilon_K \to \infty$ the model collapses to the baseline. We set $\epsilon_K = 4$, the smallest integer value at which the model has a unique stable equilibrium under our parameter values; House and Shapiro (2008) estimate values between about 6 and 14, so this is a strong version of the friction, and Section 7 reports results for larger values.

2.6 Equilibrium

Definition (Competitive Rational Expectations Equilibrium). Given exogenous shock processes $\{z_t, d_t, \mu_t\}_{t=0}^\infty$ and initial conditions $(k_{-1},\, k_{-1}^\tau,\, s_{C,-1},\, s_{G,-1},\, b_{-1})$, a *competitive rational expectations equilibrium* is a collection of sequences of quantities

$$\left\{c_t,\, i_t,\, g_t,\, x_t,\, y_t,\, h_t,\, u_t,\, k_t,\, k_t^\tau,\, b_t,\, \mathrm{div}_t,\, s_{C,t},\, s_{G,t},\, x_{c,t},\, x_{g,t}\right\}_{t=0}^\infty,$$

prices and shadow values

$$\left\{w_t,\, r_t^k,\, R_t,\, \pi_t,\, \pi_t^w,\, mc_t,\, q_t,\, \lambda_t,\, v_{\tau,t},\, \tilde{w}_t,\, \tilde{\nu}_t^c,\, \tilde{\nu}_t^g,\, \tilde{\nu}_t^i\right\}_{t=0}^\infty,$$

and fiscal instrument sequences $\{\tau_t^c,\, \tau_t^w,\, \tau_t^k,\, \tau_t^i,\, \tau_t^d,\, \tau_t^{ic},\, e_t^\tau\}_{t=0}^\infty$ such that:

(i) Taking prices and tax rates as given, the representative household maximizes (2.1) subject to the budget constraint (2.6), capital accumulation (2.7), and tax basis accumulation (2.8), yielding the first-order conditions (2.9)–(2.15).

(ii) Taking wages w_t and rental rates r_t^k as given, each intermediate goods firm $i \in [0,1]$ minimizes costs, yielding factor demands (2.20)–(2.21), and sets its nominal price to maximize the present discounted value of dividends subject to Rotemberg adjustment costs and the deep habits demand structure (2.16)–(2.17), yielding customer-base shadow values (2.23)–(2.25) and the price Phillips curve (2.26).

(iii) Each household sets its nominal wage to maximize utility subject to Rotemberg wage adjustment costs, yielding the wage Phillips curve (2.28).

(iv) The government satisfies its budget constraint (2.31) with tax revenue defined by (2.32), and each fiscal instrument follows its policy rule (2.33)–(2.40).

(v) The central bank sets the gross nominal interest rate R_t according to the Taylor rule (2.42).

(vi) Labor, capital, bond, and goods markets clear according to conditions (2.45)–(2.47).

(vii) Expectations $E_t[\cdot]$ are rational: agents' beliefs are consistent with the probability distribution implied by the model.

2.6.1 Market Clearing

In a symmetric equilibrium all intermediate goods firms make identical factor choices, so labor and capital markets clear when household factor supply equals aggregate firm demand:

$$h_t = \int_0^1 h_{i,t}\, di \tag{2.45}$$

$$k_{t-1} = \int_0^1 k_{i,t-1}\, di \tag{2.46}$$

In symmetric equilibrium these reduce to $h_{i,t} = h_t$ and $k_{i,t-1} = k_{t-1}$ for all $i \in [0,1]$.

The bond market clears when household bond holdings equal the government's outstanding debt. Since both the household budget constraint (2.6) and the government budget constraint (2.31) use the same variable b_t, this condition holds by construction.

Goods market clearing requires total net output to equal total absorption. Combining the output definition (2.22) with the resource constraint (2.30) yields:

$$y_t = c_t + g_t + i_t + \frac{\gamma_1}{1+\sigma_u}\left(u_t^{1+\sigma_u} - 1\right) k_{t-1} + \frac{\alpha_W}{2}(\pi_t^w - \bar{\pi})^2 w_t \tag{2.47}$$

Output y_t is already net of the Rotemberg price adjustment cost (see equation (2.22)), so only the utilization cost and wage adjustment cost appear on the right-hand side. By Walras' law, the bond market clearing condition is redundant given (2.45), (2.46), and (2.47).

2.7 Solving the Model

The equilibrium conditions described in Section 2 form a system of 46 nonlinear expectational difference equations in 46 structural endogenous variables, driven by 13 exogenous shock processes. The model is solved and estimated in Dynare using a three-step procedure: analytical computation of the deterministic steady state, first-order perturbation approximation of the equilibrium system, and Bayesian estimation via the Kalman filter and Metropolis-Hastings algorithm.

2.7.1 Steady State

The model's deterministic steady state is computed analytically by setting all shocks to their unconditional means ($z = d = \mu = 1$; all fiscal innovations zero) and imposing stationarity. The calibrated parameters in Table 1 jointly pin down the steady-state tax rates and the aggregate ratios $\bar{g}/\bar{y}$ and $\bar{b}/\bar{y}$, from which all remaining steady-state values

follow in closed form. In particular, the steady-state rental rate $\bar{r}^k$, wage $\bar{w}$, output $\bar{y}$, consumption $\bar{c}$, investment $\bar{\imath}$, and government debt $\bar{b}$ are each expressed as analytic functions of the calibrated parameters. The fixed operating cost ψ is then set residually to match the target dividend share $\overline{\text{div}}/\bar{y} = 0.05$. Because several estimated parameters (deep habit parameters, risk aversion, Rotemberg costs) enter the steady-state expressions, all steady-state values are recomputed at each draw of the Metropolis-Hastings algorithm.

2.7.2 First-Order Perturbation Approximation

Let y_t denote the n_y-vector of endogenous variables and $u_t \sim \mathcal{N}(0, \Sigma)$ the n_u-vector of structural shocks. The equilibrium conditions can be written compactly as:

$$E_t[f(y_{t+1},\, y_t,\, y_{t-1},\, u_t;\, \Theta)] = 0 \tag{2.48}$$

where Θ is the structural parameter vector. Dynare takes a first-order Taylor expansion of (2.48) around the deterministic steady state $(\bar{y},\, \mathbf{0})$, yielding the linear policy function:

$$\hat{y}_t = A(\Theta)\, \hat{y}_{t-1} + B(\Theta)\, u_t \tag{2.49}$$

where $\hat{y}_t \equiv y_t - \bar{y}$ denotes deviations from steady state and the coefficient matrices A and B are functions of the structural parameters. Dynare computes A and B using the generalized Schur (QZ) decomposition Klein (2000), which handles both predetermined and forward-looking variables. A unique stationary rational expectations solution exists if and only if the number of eigenvalues of A outside the unit circle equals the number of non-predetermined (forward-looking) variables, the Blanchard-Kahn conditions Blanchard and Kahn (1980). Dynare verifies these conditions numerically at the posterior mode and at every point during estimation at which the likelihood is evaluated.

By Walras' law, one market clearing condition is redundant; Dynare drops the goods market clearing equation (2.47) from the system prior to solving, leaving 45 independent structural equations. The policy function (2.49) therefore maps the predetermined state variables (the capital stock and its tax basis, government debt, the two habit stocks, the lagged fiscal instruments, and lagged output, investment, the interest rate, and the real wage) and the 13 structural shocks into the full set of endogenous variables.

2.7.3 Likelihood Evaluation and Bayesian Estimation

The policy function (2.49) together with the 12 measurement equations (Section 2.8) form a linear Gaussian state-space system. The Kalman filter recursively evaluates the prediction-error decomposition of the log-likelihood:

$$\log \mathcal{L}(Z^T \mid \Theta) = -\frac{T \cdot n_{obs}}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^{T} \left[\log |F_t| + \nu_t' F_t^{-1} \nu_t \right] \tag{2.50}$$

where $\nu_t = z_t^{obs} - E_{t-1}[z_t^{obs}]$ is the n_{obs}-vector of one-step-ahead prediction errors and F_t is its conditional variance. The Kalman filter is initialized with a diffuse prior on the state vector. Observed data are demeaned prior to estimation to handle potential mismatches between model steady states and sample means.

Combining the likelihood with the prior distributions specified in Table 2 yields the log-posterior kernel:

$$\log p(\Theta \mid Z^T) = \log \mathcal{L}(Z^T \mid \Theta) + \log p(\Theta) + \text{const.} \tag{2.51}$$

The posterior mode is located with Dynare's Newton-type optimizer (`mode_compute = 5`). We sample the posterior with a Gaussian random-walk Metropolis-Hastings algorithm whose proposal covariance is taken from the posterior draws of a preliminary run rather than from the inverse Hessian at the mode, which in our experience gives a better-shaped proposal for a posterior of this size. We run four independent chains of 1,000,000 draws each and discard the first half of each chain as burn-in, leaving 2,000,000 retained draws. We judge convergence by the potential scale reduction factor of Gelman and Rubin (1992) for every estimated parameter, the drift of the log posterior over the retained half of each chain, the agreement of the four chains, and the effective sample size of every parameter. For the baseline model the largest potential scale reduction factor across the 49 estimated parameters is 1.005, and every estimated model in the paper has a largest factor below 1.02. Impulse response functions and fiscal multipliers are computed from 1,000 draws taken at random from the retained posterior sample.

2.8 Calibration and Estimation

The model's parameter vector is partitioned into two groups. A subset of parameters is *calibrated*: fixed prior to estimation at values drawn from the literature or from long-run data targets, because they are weakly identified by the business-cycle observables used in estimation. The remaining parameters are *estimated* via Bayesian methods using U.S. quarterly data, following Zubairy (2014) and Smets and Wouters (2003), as described in Section 2.7.

2.8.1 Calibrated Parameters

Table 1 reports the calibrated parameters. The discount factor β is set to imply a 3% annual real interest rate. The quarterly economic depreciation rate $\delta = 0.025$ corresponds to a standard 10% annual rate. The capital income share $\theta = 0.30$ matches the average labor share of income in U.S. national accounts. The price elasticity of substitution $\eta = 5.3$ implies a steady-state markup of roughly 23%, while the labor variety elasticity $\eta_w = 21.0$ is calibrated to match a 5% steady-state wage markup consistent with the Rotemberg wage-setting framework.

Four parameters are fixed before estimation because the data appear to say little about them. In a preliminary estimation with all four free, the posterior standard deviation was about as large as the prior standard deviation for the private habit persistence θ^c and the wage adjustment cost α_W, and only somewhat smaller for the utilization curvature σ_u and the government habit depth b_g, and these parameters traded off against each other along flat ridges of the posterior. We set $\theta^c = 0.53$, the posterior mean in Zubairy (2014); $\alpha_W = 100$, which corresponds to wage contracts lasting about four quarters under the usual mapping between Rotemberg and Calvo adjustment costs and equals the prior mean in Zubairy (2014); $\sigma_u = 2.5$, a conventional value that also equals her prior mean; and $b_g = 0.70$, her prior mean, which is close to the posterior median of our preliminary estimation. The multipliers barely move with α_W and move by roughly ten percent with θ^c over its posterior range. The capital tax multiplier depends on σ_u and the spending multiplier on b_g, so Section 7 reports both across a range of values. The same values are used in every model we estimate.

Steady-state hours worked $\bar{h} = 0.50$ normalizes the time endowment to unity and follows Zubairy (2014). The government spending share $\bar{g}/\bar{y} = 0.21$ and debt-to-output ratio

$\bar{b}/\bar{y} = 0.32$ are set to the sample averages over 1961Q1–2013Q4. Steady-state inflation is normalized to $\bar{\pi} = 1$ (zero net inflation).

Steady-state tax rates are calibrated to the sample averages over the same period. The capital income tax $\bar{\tau}^k = 0.42$ and labor income tax $\bar{\tau}^w = 0.22$ follow the effective tax rate construction of Jones (2002). The consumption tax $\bar{\tau}^c = 0.06$ follows the construction of Mendoza et al. (1994) and reflects the effective tax rate on private consumption expenditures. The interest income and dividend tax rates $\bar{\tau}^i = 0.25$ and $\bar{\tau}^d = 0.30$ are the sample averages of the interest and qualified dividend marginal tax rates from the McGrattan TAXSIM series; the interest income tax rate stays at this value throughout. The investment tax credit rate $\bar{\tau}^{ic} = 0.02$ and expensing rate $\bar{e}^\tau = 0.06$ are sample averages of the House and Shapiro (2008) ITC series and the BEA asset-weighted bonus depreciation series, respectively. The tax depreciation rate $\delta^\tau = 0.025$ equals the economic depreciation rate as a baseline. The dividend share of output $\overline{\text{div}}/\bar{y} = 0.05$ is set to the sample average; the fixed operating cost ψ in the production function is determined residually to hit this target in steady state.

2.8.2 Data

We use quarterly U.S. data from 1961Q1 to 2013Q4. Real per-capita consumption, investment, government purchases, and gross domestic product are constructed from BEA National Income and Product Accounts and the U.S. Census population series, then log-linear-detrended and expressed as percent deviations from trend. Real federal debt is the Dallas Fed market-value series (MVPHGFD027MNFRBDAL). Inflation is measured as gross GDP deflator inflation, and the interest rate is the Federal Funds Rate.

Effective tax rates on labor and capital income follow the NIPA-based construction of Jones (2002), and the consumption tax rate follows Mendoza et al. (1994). The dividend tax rate is taken from the McGrattan TAXSIM series (average marginal rate on qualified dividends), and the interest income tax rate is the corresponding TAXSIM series on interest income. The investment tax credit rate is constructed following House and Shapiro (2008) as a weighted aggregation of the statutory ITC schedule across equipment types. The expensing rate is the BEA asset-weighted average of statutory bonus depreciation provisions. Sample coverage is dictated by the availability of the TAXSIM series, which begins in 1960 and ends in 2013. All tax rate series are constructed as fractional deviations from their sample means.

Two features of the data motivate specific modeling choices. First, the 2008Q4 observation of market-value federal debt exhibits a large jump driven by valuation and issuance movements that are not modeled explicitly; the debt residual ε_t^b in the government budget constraint absorbs this so that the fiscal rules do not reconcile the jump through counterfactual movements in the tax instruments. Second, the ITC rate was inactive for most quarters after 1986 and bonus depreciation was inactive prior to 2001, so both τ_t^{ic} and e_t^τ take values at or near zero for extended periods of the sample. This is the reason the corresponding fiscal rules use the level-deviation form (equations 2.39 and 2.40) rather than the log-ratio form used for the other tax instruments.

2.8.3 Bayesian Estimation

The remaining 49 parameters of the baseline model (36 structural and fiscal-rule parameters and 13 shock standard deviations) are estimated jointly using Bayesian techniques; the

model with AR(1) fiscal rules has 34. We match 12 quarterly observables from U.S. data to 12 of the model's measurement equations:

1. $\hat{c}_t = 100(c_t - \bar{c})/\bar{c}$: private consumption (percent deviation from SS)
2. $\hat{\imath}_t = 100(i_t - \bar{\imath})/\bar{\imath}$: private investment (percent deviation)
3. $\hat{g}_t = 100(g_t - \bar{g})/\bar{g}$: government purchases (percent deviation)
4. $\pi_t - \bar{\pi}$: gross price inflation (level deviation)
5. $R_t - \bar{R}$: gross nominal interest rate (level deviation)
6. $(\tau_t^k - \bar{\tau}^k)/\bar{\tau}^k$: capital income tax rate (fractional deviation)
7. $(\tau_t^w - \bar{\tau}^w)/\bar{\tau}^w$: labor income tax rate (fractional deviation)
8. $100(b_t - \bar{b})/\bar{b}$: government debt (percent deviation)
9. $(\tau_t^c - \bar{\tau}^c)/\bar{\tau}^c$: consumption tax rate (fractional deviation)
10. $(\tau_t^d - \bar{\tau}^d)/\bar{\tau}^d$: dividend tax rate (fractional deviation)
11. $\tau_t^{ic} - \bar{\tau}^{ic}$: investment tax credit rate (level deviation)
12. $e_t^\tau - \bar{e}^\tau$: expensing rate (level deviation)

Transfer payments x_t are excluded from the observable set because the 2008–2010 ARRA stimulus produced outliers of a magnitude the linearized model cannot fit; without this exclusion, the transfer shock variance inflates and distorts other fiscal-rule estimates. The interest income tax rate τ_t^i is also excluded because it has a 0.77 correlation with the dividend tax rate at the quarterly frequency, both being constructed from the same TAXSIM series; including both induces collinearity in the measurement equations. Since it is not observed, the interest income tax rate is held constant (Section 2.3). The transfer shock ε_t^x is retained as an unobserved structural shock identified through cross-equation restrictions. Together with the debt-residual shock ε_t^b, the model has 13 structural shocks and 12 observables. Observed data are demeaned prior to estimation (`prefilter = 1`) to accommodate potential mismatches between model steady states and sample means.

Table 2 reports the prior distributions and posterior estimates of the baseline model. Prior choices follow Zubairy (2014) for the parameters common to both models. The deep habit parameters b_c and θ^g receive Beta priors. The risk aversion coefficient γ, the investment adjustment cost κ, and the price adjustment cost α_P receive Normal priors. The persistence parameters of the fiscal rules and exogenous processes receive Beta$(0.7, 0.2)$ priors, except for government purchases, whose prior mean is 0.8. The debt-feedback coefficients of the capital, labor, consumption, and dividend tax rules and of the spending and transfer rules receive Gamma$(0.5, 0.25)$ priors, so that taxes rise and spending and transfers fall when debt is above steady state; for the investment tax credit and the expensing rate, whose direction of response to debt is not obvious, the prior is Normal$(0, 0.25)$. Output-feedback coefficients follow Zubairy (2014) for government purchases, transfers, and the capital and labor taxes, and receive Normal$(0, 0.1)$ priors for the other instruments. All shock standard deviations receive inverse-Gamma priors. The twelve fiscal-rule parameters for the consumption tax, dividend tax, investment tax credit, and expensing rate, and the four corresponding shock standard deviations, as well as the standard deviation of the debt residual, are new to this paper.

3 Results

3.1 Computing the Multipliers

For each fiscal instrument j, we compute the present-value multiplier at horizon H as the discounted cumulative change in output per dollar of discounted cumulative change in the instrument's own revenue or outlay:

$$M_j(H) = s_j \frac{\sum_{t=0}^{H-1} \bar{R}^{-t} \, \Delta y_t^{(j)}}{\sum_{t=0}^{H-1} \bar{R}^{-t} \, \Delta Rev_{j,t}^{(j)}} \tag{3.1}$$

where $\Delta y_t^{(j)}$ and $\Delta Rev_{j,t}^{(j)}$ are the responses of output and of instrument j's own revenue t quarters after a shock ε^j, $\bar{R}$ is the steady-state gross nominal interest rate, and $H = 1$ is the impact quarter. $Rev_{j,t}$ is the component of $taxrev_t$ in equation (2.32) collected by instrument j: consumption tax revenue $\tau_t^c c_t$, labor income tax revenue $\tau_t^w w_t h_t$, and capital income tax revenue for the capital tax. For the expensing rate, it is the net cost of the additional deductions relative to keeping expensing at its steady-state rate on the same investment path, $\tau_t^k \left[(e_t^\tau - \bar{e}^\tau) i_t + \delta^\tau B_{t-1} \right]$, where $B_t = (1 - \delta^\tau) B_{t-1} - (e_t^\tau - \bar{e}^\tau) i_t$ is the resulting change in the tax basis. The first term is the deduction taken immediately; the second, negative after an increase in expensing, is the revenue recovered later through smaller depreciation deductions. In the model with the capital-goods friction, $P_t^I i_t$ replaces i_t. For the investment tax credit, it is the outlay $\tau_t^{ic} i_t$; and for government purchases, it is g_t. The sign s_j is chosen so that the multiplier is positive when output moves in the expansionary direction of the policy, a tax cut or an increase in spending, the credit, or expensing. Because the model is solved to first order, the multiplier does not depend on the size of the shock.

Under the baseline fiscal rules, in some draws total tax revenue stops falling within the horizons we report, so that a tax cut or investment incentive is self-financing: the present value of output rises while the present value of total tax revenue does not fall. We keep the own-revenue multiplier in equation (3.1) for these draws, since it still measures output per dollar of the instrument's own fiscal cost, and we report separately the share of posterior draws in which the policy is self-financing at each horizon. Spending and transfers are excluded from this measure.

In the capital tax experiment we set the coefficient ϕ on ε_t^k in the labor tax rule (2.36) to zero, so that a capital tax shock does not also move the labor tax rate on impact and the experiment changes a single instrument. This affects only the simulation, not the estimation.

We use the instrument's own revenue rather than total tax revenue as the denominator for two reasons. First, the instruments differ greatly in the size of their bases, and the own-revenue denominator measures output per dollar of the instrument's direct fiscal cost, which is closer to how such policies are scored. Second, under fiscal rules in which every tax responds to debt and output, the change in total tax revenue includes the responses of all the other instruments, and for the capital tax and the investment tax credit its present value changes sign within the first five years, so the total-revenue multiplier becomes very large and changes sign at the horizon where its denominator passes through zero. A total-revenue denominator is therefore not informative in our baseline model, although for government spending it gives the same values as ours by construction.

Point estimates are posterior medians, and intervals in brackets are 5th and 95th percentiles across 1,000 draws taken at random from the retained posterior sample.

We do not report multipliers for lump-sum transfers or the dividend tax. In this representative-agent model, households are Ricardian and a transfer has no direct effect on their choices, and the dividend tax falls on pure profits and does not enter any first-order condition. Any output response to either instrument therefore comes only from how the government finances the resulting change in debt: under AR(1) fiscal rules both multipliers are zero at every horizon, and under the baseline rules they measure the effect of the other instruments' responses to debt rather than of the instrument itself.

3.2 Short-Run Multipliers

Table 3 reports the multipliers of the baseline model at horizons of 1, 4, 12, and 20 quarters. Panel A gives our estimates for the six instruments, and Panel B reproduces the corresponding values from Zubairy (2014), Table 2.

On impact, government spending has the largest multiplier, 1.12, close to the 1.07 in Zubairy (2014). The capital income tax follows at 1.35 per dollar of capital tax revenue, and the investment tax credit at 0.76. The labor tax (0.32) and the consumption tax (0.27) have similar and much smaller impact multipliers, and the expensing rate has the smallest, 0.20. Our tax multipliers are larger than those in Panel B, but the two are not directly comparable, since her denominator is the change in total tax revenue and the two models differ in their tax instruments and calibration.

Within the first year the ranking changes. By four quarters the capital tax multiplier has reached 2.13 and the investment tax credit 1.54, both above the spending multiplier of 1.04. The labor tax multiplier rises to 0.62, and the consumption tax and expensing multipliers to 0.39 and 0.37. The instruments that act directly on the cost of capital, the capital income tax and the investment tax credit, therefore appear to be the most effective per dollar of forgone revenue at all horizons beyond the impact quarter.

The expensing multiplier is small relative to that of the investment tax credit, although both lower the cost of investment. A higher expensing rate mostly moves deductions forward in time: it lowers capital tax revenue sharply in the quarter the investment is made, while the investment incentive it creates is only the value of receiving the deduction earlier. Per dollar of near-term revenue cost, the incentive therefore appears to be much smaller than for a credit of the same cost.

Figures 3 through 8 plot the posterior impulse responses of output, consumption, investment, hours, inflation, and the interest rate to a one-standard-deviation expansionary shock to each instrument: an increase in spending, the investment tax credit, or the expensing rate, or a cut in a tax rate. Solid lines are posterior medians and shaded regions are 5–95 percentile bands. Output, consumption, investment, and hours are in percent deviations from steady state; inflation and the interest rate are in annualized percentage points.

3.3 Multipliers over the Medium Term

Figure 1 plots the multipliers of the baseline model as a function of the horizon up to 40 quarters. The capital tax and investment tax credit multipliers stay large: at 20 quarters they are 2.70 and 2.14, close to their 12-quarter values, and at 40 quarters 2.72 and 2.64. The other multipliers decline after the first year. The spending multiplier falls from 1.04 at four quarters to 0.42 at twelve and -0.26 at twenty, and the labor tax multiplier peaks at 0.76 after two years and falls to 0.18 at twenty quarters. The consumption tax and expensing multipliers are close to zero at twelve quarters and negative at twenty, at -0.62

and -0.34, and the 90 percent intervals exclude zero for both.

These negative multipliers at longer horizons arise under fiscal rules in which every tax responds to debt and output. A tax cut raises debt, and the debt is repaid partly through higher distortionary taxes and lower government purchases in later years, which may well reduce output by more than the original cut raised it. Section 4 shows that under fiscal rules in which only transfers respond to debt, none of the multipliers turns negative.

Although the capital tax and investment tax credit multipliers remain finite, by twenty quarters total tax revenue no longer falls for these two instruments in most draws: the policy is self-financing in 74 percent of draws for the capital tax and 73 percent for the investment tax credit, against 16 and 12 percent at twelve quarters (Table 3). This seems to reflect in part the fiscal rules, since the other tax rates rise as debt rises. Under the AR(1) rules, in which only transfers respond to debt, no draw is self-financing for any instrument.

3.4 The Small Consumption Tax Multiplier

The consumption tax multiplier is small at every horizon: 0.27 on impact, 0.39 after a year, and close to zero after three years. This seems to reflect the limited ways in which a consumption tax can affect a representative household.

First, the consumption tax works mainly through the same margin as the labor income tax. The labor supply condition depends on the real wage measured in consumption units, so the relevant wedge is $(1 - \tau_t^w)/(1 + \tau_t^c)$, and a cut in the consumption tax raises the return to work in the same way as a cut in the labor tax. The two impact multipliers are indeed similar, 0.27 and 0.32. Second, a temporary cut makes consumption today cheaper relative to consumption in the future, which shifts spending toward the present. With an estimated persistence of the consumption tax rule of 0.92, a cut is expected to last a while, which weakens this intertemporal effect.

Third, households in the model are Ricardian. A consumption tax cut does not raise consumption by raising disposable income, since households anticipate the taxes needed to repay the debt. This is the channel that seems to make consumption tax cuts effective in models with hand-to-mouth households, and it is absent here. Deep habits in consumption further slow the response of consumption to its relative price; the posterior median of the habit depth b_c is 0.76.

Under the baseline rules the consumption tax multiplier turns negative after about three years. The debt created by the cut raises the other distortionary taxes and, through the debt feedback in its own rule, eventually raises the consumption tax itself above its steady state. Under AR(1) fiscal rules the multiplier rises to about one third and stays there (Table 4). A larger consumption tax multiplier would probably require a share of hand-to-mouth households, the more common route in the recent literature.

4 Sensitivity to Government Financing

The baseline fiscal rules let every instrument respond to debt, so the debt created by a tax cut is repaid through a mix of higher distortionary taxes, lower spending, and lower transfers. To see how much the financing matters, we estimate the model again with AR(1) fiscal rules in which only lump-sum transfers respond to debt (equation 2.41). Since households are Ricardian, financing through transfers is neutral, and the multipliers under these rules measure the effect of each instrument without the effect of its financing.

Table 4 and Figure 2 compare the multipliers across specifications. In the first year the baseline rules give larger multipliers for every tax instrument; at four quarters, for instance, the labor tax multiplier is 0.62 under the baseline rules and 0.41 under AR(1) rules, and the investment tax credit multiplier 1.54 and 0.87. After that the ranking reverses for most instruments. Under AR(1) rules every multiplier is positive at twenty quarters: 0.57 for spending, 0.83 for the labor tax, 3.02 for the capital tax, 0.33 for the consumption tax, 1.77 for the investment tax credit, and 0.45 for expensing. Under the baseline rules the spending, consumption tax, and expensing multipliers are negative at twenty quarters, and the labor tax multiplier is 0.18. This pattern seems consistent with the view that the negative long-horizon multipliers in the baseline model come from repaying debt through distortionary taxes rather than from the instruments themselves. Anticipated future tax increases may also explain the larger short-run multipliers under the baseline rules, since they give households and firms a reason to bring work and investment forward. Because the two models are estimated separately, however, the comparison also reflects differences in the posteriors; switching off the feedback terms on the same posterior draws would isolate the financing channel more cleanly.

The capital tax multiplier is the least sensitive to the fiscal rules: it is between 2.7 and 3.0 at twenty quarters in all four U.S. specifications. The investment tax credit is the one instrument whose multiplier is larger under the baseline rules, 2.23 against 1.43 at twelve quarters. This may reflect the estimated response of the credit itself to debt and output, which the AR(1) rules switch off.

5 Investment Incentives and the Price of Capital Goods

Comparing the separately estimated models with and without it, the capital-goods friction of Section 2.5 lowers the multipliers of the investment incentives the most. Under the baseline rules it lowers the investment tax credit multiplier from 2.23 to 1.68 at twelve quarters, about one quarter, and the expensing multiplier from 0.17 to 0.07 (Table 4). Under AR(1) rules it lowers the investment tax credit multiplier from 1.43 to 1.19. The labor tax multiplier falls by about 15 percent, the capital tax multiplier by about five percent, and the spending multiplier is slightly higher with the friction.

The friction works through incidence rather than through the fiscal cost. Comparing the friction model under the baseline rules with the same model at $\epsilon_K = \infty$, on the same posterior draws, the present value of the revenue cost of the investment tax credit changes by less than four percent at every horizon, while the present value of its output response is 15 to 40 percent lower. A temporary subsidy to investment raises the price of capital goods, so part of it is captured by the capital-goods sector instead of raising the quantity of investment, as House and Shapiro (2008) find for bonus depreciation. Table 5 shows that the effect shrinks as the supply elasticity rises toward the range estimated by House and Shapiro (2008): at $\epsilon_K = 9$ the investment tax credit multiplier at twelve quarters is 2.13. The estimates for the two models with the friction are in Table 6.

6 Estimates for the United Kingdom

We also estimate the model with AR(1) fiscal rules on UK data for 1997Q2–2016Q4. Fewer UK tax series are available, so the UK model has nine observables, no investment tax credit, and steady-state expensing, interest, and dividend tax rates set at their U.S. values

(Appendix A-2). The UK multipliers are broadly similar to the U.S. estimates under the same fiscal rules (Table 7 and Figure 9). At twelve quarters the spending multiplier is 0.68 against 0.70 in the United States, the labor tax 0.61 against 0.72, the capital tax 2.48 against 2.82, and the consumption tax 0.38 against 0.34. The consumption tax multiplier is larger on impact in the United Kingdom, 0.30 against 0.17. The expensing multiplier is more than twice as large, 0.73 against 0.32, which may partly reflect the higher UK capital tax rate, but it also rests on the U.S. steady-state expensing rate and should be read with caution.

7 Sensitivity to Calibrated Parameters

Table 5 reports how the multipliers change with the calibrated parameters that matter most for them. Each entry re-solves the model on the same 1,000 posterior draws with one parameter changed, so it shows the effect of the parameter conditional on the posterior rather than the result of a new estimation.

The capital tax multiplier depends on the curvature of the utilization cost. At twelve quarters it is 3.33 with $\sigma_u = 2.0$, 2.82 at the baseline value of 2.5, and 2.46 with $\sigma_u = 3.0$. A flatter utilization cost lets firms raise capital services quickly when the after-tax return to capital rises, which seems to be why the capital tax multiplier is large on impact. The spending multiplier depends on the depth of the government consumption habit: at twelve quarters it is 0.31 with $b_g = 0.55$, 0.42 at the baseline value of 0.70, and 0.70 with $b_g = 0.85$. In both cases the ranking of the instruments is unchanged.

Panel B reports the multipliers of the model with the capital-goods friction as the supply elasticity ϵ_K rises from the baseline value of 4 through the range of 6 to 14 estimated by House and Shapiro (2008), and with the friction switched off. The investment tax credit multiplier at twelve quarters rises from 1.68 to 2.29 at $\epsilon_K = 14$ and 2.63 without the friction. Every multiplier rises with ϵ_K, so the friction is not specific to purchase subsidies: over the same range the labor tax multiplier rises from 0.55 to 0.94, the capital tax multiplier from 2.69 to 3.22, and the spending multiplier from 0.50 to 0.65. These values at $\epsilon_K = \infty$ differ from the estimates of the baseline model because they use the posterior of the friction model.

8 Conclusion

TBD

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Tables

Table 1: Calibrated Parameters

Parameter	Description	Value
Preferences and technology		
β	Discount factor	$1/(1.03)^{1/4}$
δ	Quarterly economic depreciation rate	0.025
θ	Capital income share	0.30
η	Elasticity of substitution, goods varieties	5.3
η_w	Elasticity of substitution, labor varieties	21.0
Weakly identified parameters, fixed before estimation		
θ^c	Private habit stock persistence	0.53
b_g	Government consumption habit depth	0.70
σ_u	Utilization cost curvature	2.5
α_W	Rotemberg wage adjustment cost	100
Steady-state targets		
$\bar{\pi}$	Steady-state gross inflation	1.0
$\bar{h}$	Steady-state hours worked	0.50
$\bar{g}/\bar{y}$	Government spending share of output	0.21
$\bar{b}/\bar{y}$	Debt-to-output ratio	0.32
$\overline{\text{div}}/\bar{y}$	Dividend share of output	0.05
Steady-state tax rates		
$\bar{\tau}^k$	Capital income tax rate	0.42
$\bar{\tau}^w$	Labor income tax rate	0.22
$\bar{\tau}^c$	Consumption tax rate	0.06
$\bar{\tau}^i$	Interest income tax rate (held constant)	0.25
$\bar{\tau}^d$	Dividend tax rate	0.30
$\bar{\tau}^{ic}$	Investment tax credit rate	0.02
$\bar{e}^\tau$	Immediate expensing rate	0.06
δ^τ	Tax depreciation rate	0.025
Capital-goods friction (extension only)		
ϵ_K	Supply elasticity of capital goods	4

Note: U.S. sample averages over 1961Q1–2013Q4 for the steady-state targets and tax rates. The four weakly identified parameters are fixed at the same values in every model; Section 2.8 gives the sources and Table 5 the sensitivity of the results to σ_u and b_g.

Table 2: Prior Distributions and Posterior Estimates, Baseline Model

Parameter	Description	Prior	Mean	Std	Median	5%	95%
Deep habits							
b_c	Private consumption habit depth	Beta	0.700	0.100	0.764	0.691	0.829
θ^g	Government habit stock persistence	Beta	0.800	0.100	0.861	0.699	0.949
Structural parameters							
γ	Risk aversion	Normal	2.000	1.000	2.047	1.606	2.506
κ	Investment adjustment cost	Normal	2.000	0.500	3.121	2.510	3.760
α_P	Rotemberg price adjustment cost	Normal	17.0	5.000	33.8	28.2	39.9
Monetary policy							
α_R	Interest rate smoothing	Beta	0.800	0.200	0.700	0.651	0.744
α_π	Inflation response	Normal	1.600	0.200	1.986	1.830	2.172
α_Y	Output response	Normal	0.100	0.050	0.014	−0.004	0.034
Exogenous processes							
ρ_z	Technology	Beta	0.700	0.200	0.991	0.983	0.997
ρ_d	Preference	Beta	0.700	0.200	0.664	0.581	0.745
ρ_μ	Investment efficiency	Beta	0.700	0.200	0.530	0.437	0.620
Fiscal rules: persistence							
ρ_g	Government purchases	Beta	0.800	0.200	0.951	0.928	0.973
ρ_x	Transfers	Beta	0.700	0.200	0.984	0.957	0.998
ρ_k	Capital tax	Beta	0.700	0.200	0.960	0.940	0.980
ρ_w	Labor tax	Beta	0.700	0.200	0.973	0.950	0.991
ρ_c	Consumption tax	Beta	0.700	0.200	0.915	0.874	0.954
ρ_d^τ	Dividend tax	Beta	0.700	0.200	0.961	0.919	0.991
ρ_{ic}	Investment tax credit	Beta	0.700	0.200	0.858	0.814	0.901
ρ_e	Expensing rate	Beta	0.700	0.200	0.951	0.904	0.989
Fiscal rules: debt feedback							
$\rho_{g,b}$	Government purchases	Gamma	0.500	0.250	0.005	0.002	0.008
$\rho_{x,b}$	Transfers	Gamma	0.500	0.250	0.138	0.063	0.240
$\rho_{k,b}$	Capital tax	Gamma	0.500	0.250	0.029	0.020	0.038
$\rho_{w,b}$	Labor tax	Gamma	0.500	0.250	0.017	0.009	0.026
$\rho_{c,b}$	Consumption tax	Gamma	0.500	0.250	0.009	0.005	0.014
$\rho_{d,b}$	Dividend tax	Gamma	0.500	0.250	0.031	0.015	0.053
$\rho_{ic,b}$	Investment tax credit	Normal	0.000	0.250	−0.010	−0.013	−0.006
$\rho_{e,b}$	Expensing rate	Normal	0.000	0.250	−0.005	−0.024	0.015
Fiscal rules: output feedback							
$\rho_{g,y}$	Government purchases	Normal	−0.050	0.050	0.044	0.014	0.074
$\rho_{x,y}$	Transfers	Normal	−0.100	0.050	−0.103	−0.183	−0.022
$\rho_{k,y}$	Capital tax	Gamma	0.150	0.100	0.070	0.028	0.120
$\rho_{w,y}$	Labor tax	Gamma	0.150	0.100	0.064	0.026	0.110
$\rho_{c,y}$	Consumption tax	Normal	0.000	0.100	−0.018	−0.055	0.019
$\rho_{d,y}$	Dividend tax	Normal	0.000	0.100	0.007	−0.102	0.110
$\rho_{ic,y}$	Investment tax credit	Normal	0.000	0.100	−0.066	−0.087	−0.044
$\rho_{e,y}$	Expensing rate	Normal	0.000	0.100	−0.004	−0.092	0.082
ϕ	Capital tax shock in labor tax rule	Normal	0.250	0.100	0.533	0.450	0.614
Shock standard deviations							
σ_g	Government purchases	Inv. Gamma	0.050	0.100	0.010	0.010	0.011
σ_x	Transfers	Inv. Gamma	0.050	0.100	0.311	0.263	0.367
σ_k	Capital tax	Inv. Gamma	0.050	0.100	0.023	0.022	0.025
σ_w	Labor tax	Inv. Gamma	0.050	0.100	0.019	0.018	0.021
σ_c	Consumption tax	Inv. Gamma	0.050	0.100	0.015	0.014	0.016
σ_d	Dividend tax	Inv. Gamma	0.050	0.100	0.056	0.052	0.061
σ_{ic}	Investment tax credit	Inv. Gamma	0.050	0.100	0.007	0.006	0.007
σ_e	Expensing rate	Inv. Gamma	0.050	0.100	0.045	0.042	0.049
σ_z	Technology	Inv. Gamma	0.050	0.100	0.010	0.009	0.012
σ_d^p	Preference	Inv. Gamma	0.050	0.100	0.024	0.019	0.032
σ_μ	Investment efficiency	Inv. Gamma	0.050	0.100	0.044	0.039	0.051
σ_m	Monetary policy	Inv. Gamma	0.050	0.100	0.006	0.006	0.007

Parameter	Description	Prior	Mean	Std	Median	5%	95%
σ_b	Debt residual	Inv. Gamma	0.050	0.100	0.007	0.007	0.008

Note: Prior mean and standard deviation. Posterior median and 5th and 95th percentiles over 2,000,000 retained draws (four chains of 1,000,000, first half of each discarded). The four weakly identified parameters θ^c, b_g, σ_u, and α_W are calibrated (Table 1).

Table 3: Fiscal Multipliers, Baseline Model

Instrument	$H = 1$	$H = 4$	$H = 12$	$H = 20$
Panel A. Baseline model (fiscal rules with debt and output feedback)				
Government spending	1.12	1.04	0.42	−0.26
	[1.05, 1.18]	[0.90, 1.15]	[0.20, 0.65]	[−0.59, 0.03]
Labor tax	0.32	0.62	0.65	0.18
	[0.26, 0.39]	[0.49, 0.76]	[0.45, 0.89]	[−0.10, 0.45]
Capital tax	1.35	2.13	2.82	2.70
	[1.23, 1.47]	[1.90, 2.36]	[2.42, 3.27]	[2.23, 3.24]
self-financing	*0%*	*0%*	*16%*	*74%*
Consumption tax	0.27	0.39	0.03	−0.62
	[0.21, 0.32]	[0.29, 0.48]	[−0.16, 0.25]	[−0.93, −0.31]
Investment tax credit	0.76	1.54	2.23	2.14
	[0.59, 0.92]	[1.23, 1.86]	[1.71, 2.81]	[1.54, 2.86]
self-financing	*0%*	*0%*	*12%*	*73%*
Expensing rate	0.20	0.37	0.17	−0.34
	[0.12, 0.27]	[0.23, 0.52]	[−0.02, 0.41]	[−0.65, −0.06]
Panel B. Zubairy (2014), Table 2				
Government spending	1.07	1.06	0.90	0.72
	[1.01, 1.13]	[0.96, 1.17]	[0.73, 1.07]	[0.49, 0.93]
Labor tax	0.13	0.32	0.68	0.85
	[0.09, 0.18]	[0.22, 0.45]	[0.41, 1.09]	[0.42, 1.58]
Capital tax	0.34	0.43	0.52	0.46
	[0.30, 0.37]	[0.34, 0.51]	[0.30, 0.73]	[0.15, 0.81]

Note: Present-value own-revenue multipliers (equation 3.1). Posterior median with 5–95 percentile intervals in brackets, from 1,000 posterior draws. A positive value means output moves in the expansionary direction of the policy. Rows labeled self-financing give the percentage of posterior draws in which the present value of total tax revenue does not fall at that horizon. It is zero for all other instruments and horizons. Sample 1961Q1–2013Q4. Panel B reproduces Zubairy (2014); her tax multipliers use the change in total tax revenue as the denominator, so they are not directly comparable to Panel A.

Table 4: Multipliers under Alternative Fiscal Rules and the Capital-Goods Friction

	$H = 12$				$H = 20$			
Instrument	FB	AR(1)	FB+K	AR(1)+K	FB	AR(1)	FB+K	AR(1)+K
Government spending	0.42	0.70	0.50	0.74	−0.26	0.57	−0.09	0.64
Labor tax	0.65	0.72	0.55	0.67	0.18	0.83	0.15	0.76
Capital tax	2.82	2.82	2.69	2.67	2.70	3.02	2.65	2.81
Consumption tax	0.03	0.34	−0.01	0.33	−0.62	0.33	−0.57	0.34
Investment tax credit	2.23	1.43	1.68	1.19	2.14	1.77	1.51	1.36
Expensing rate	0.17	0.32	0.07	0.31	−0.34	0.45	−0.45	0.39

Note: Posterior medians of the present-value own-revenue multiplier. FB: baseline fiscal rules with debt and output feedback. AR(1): only transfers respond to debt. +K: capital-goods friction with $\epsilon_K = 4$. Each column is a separately estimated model. Intervals are in Tables 3 and 6.

Table 5: Sensitivity to Calibrated Parameters

Panel A. Baseline model

	$H = 12$			$H = 20$		
	low	baseline	high	low	baseline	high
Capital tax, $\sigma_u = 2.0$ / 2.5 / 3.0	3.33	2.82	2.46	3.27	2.70	2.31
Spending, $b_g = 0.55$ / 0.70 / 0.85	0.31	0.42	0.70	−0.35	−0.26	−0.04

Panel B. Baseline fiscal rules with capital-goods friction

		Supply elasticity ϵ_K				
Instrument		4 (baseline)	6	9	14	∞
Government spending	$H = 12$	0.50	0.54	0.57	0.59	0.65
	$H = 20$	−0.09	−0.07	−0.06	−0.04	−0.02
Labor tax	$H = 12$	0.55	0.65	0.73	0.80	0.94
	$H = 20$	0.15	0.22	0.28	0.33	0.44
Capital tax	$H = 12$	2.69	2.82	2.93	3.02	3.22
	$H = 20$	2.65	2.75	2.83	2.90	3.06
Consumption tax	$H = 12$	−0.01	0.06	0.11	0.15	0.26
	$H = 20$	−0.57	−0.53	−0.50	−0.47	−0.40
Investment tax credit	$H = 12$	1.68	1.93	2.13	2.29	2.63
	$H = 20$	1.51	1.80	2.04	2.24	2.66
Expensing rate	$H = 12$	0.07	0.15	0.23	0.29	0.42
	$H = 20$	−0.45	−0.36	−0.29	−0.23	−0.10

Note: Posterior medians of the present-value own-revenue multiplier. Each entry re-solves the model on the same 1,000 posterior draws with one calibrated parameter changed; the posterior is the one estimated at the baseline calibration ($\sigma_u = 2.5$, $b_g = 0.70$, $\epsilon_K = 4$). $\epsilon_K = \infty$ switches the friction off on the posterior of the friction model, which is why it differs from Panel A.

Figures

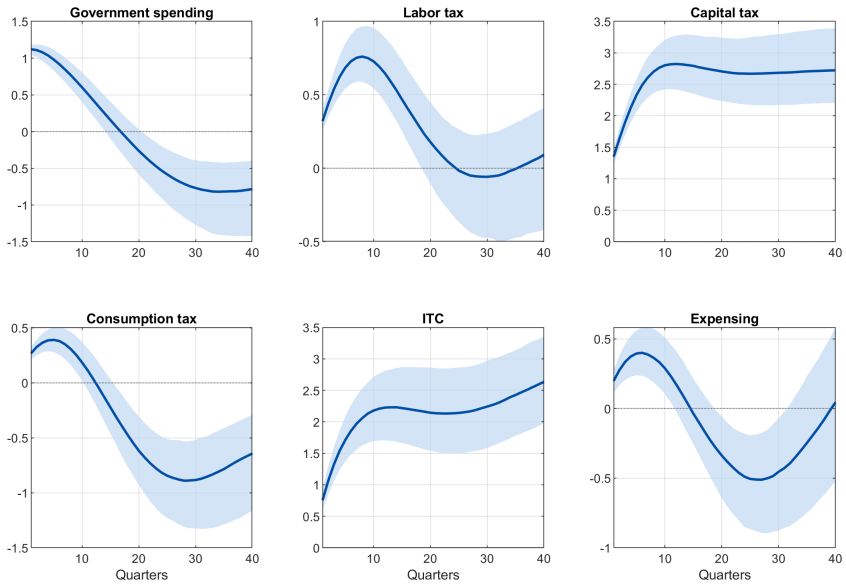


Figure 1: Multipliers by Horizon, Baseline Model
Present-value own-revenue multipliers. Posterior median and 5–95 percentile band, 1,000 posterior draws. Under the baseline fiscal rules the capital tax and the investment tax credit are self-financing in most draws by 20 quarters (Table 3).

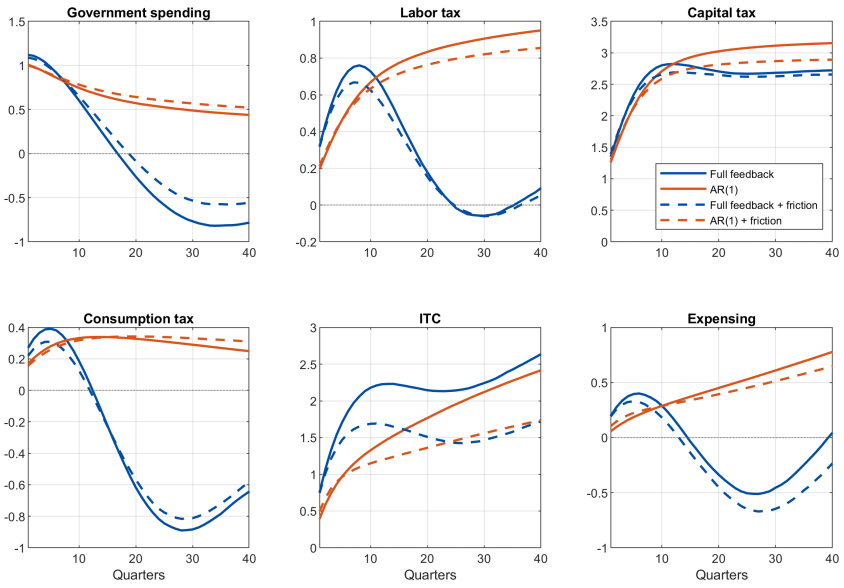


Figure 2: Multipliers by Horizon under Alternative Fiscal Rules and the Capital-Goods Friction

Posterior medians of the present-value own-revenue multiplier in the four U.S. specifications. Shares of self-financing draws are reported in Tables 3 and 6.

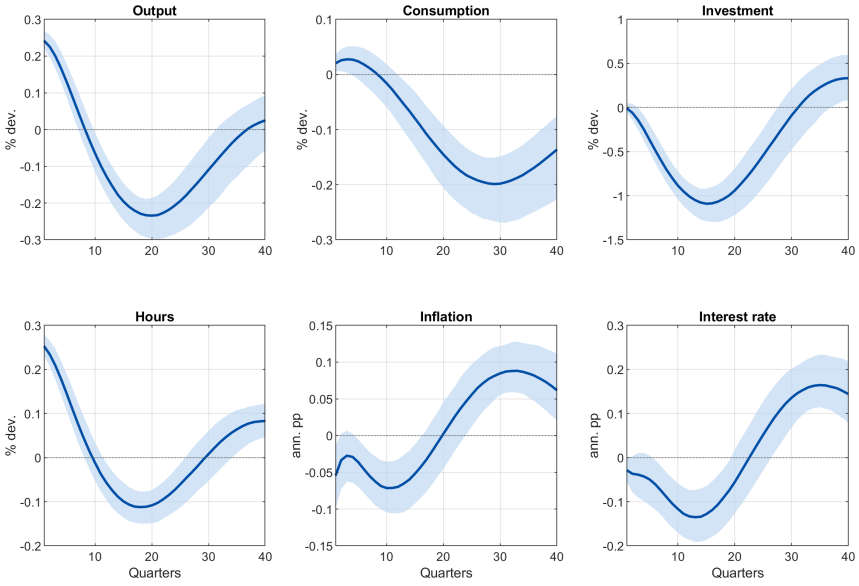


Figure 3: Impulse Responses to a Government Spending Increase

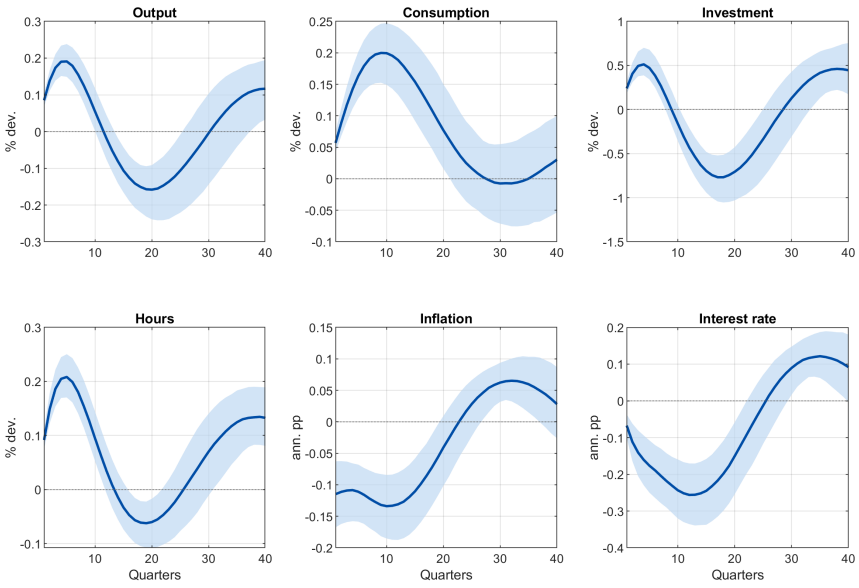


Figure 4: Impulse Responses to a Labor Income Tax Cut

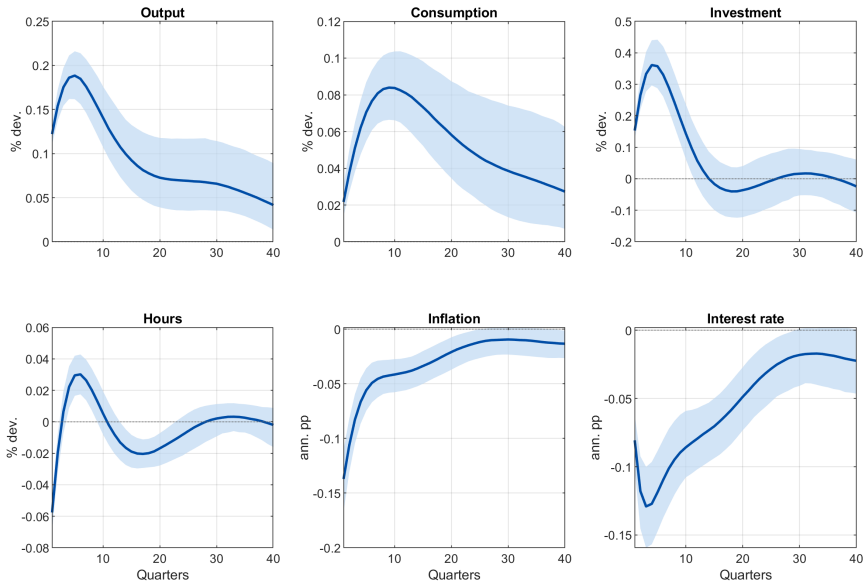


Figure 5: Impulse Responses to a Capital Income Tax Cut

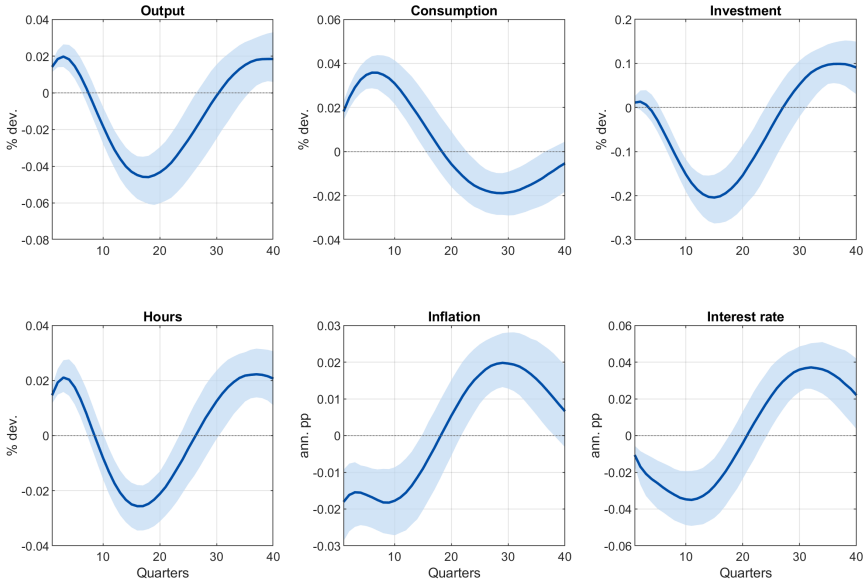


Figure 6: Impulse Responses to a Consumption Tax Cut

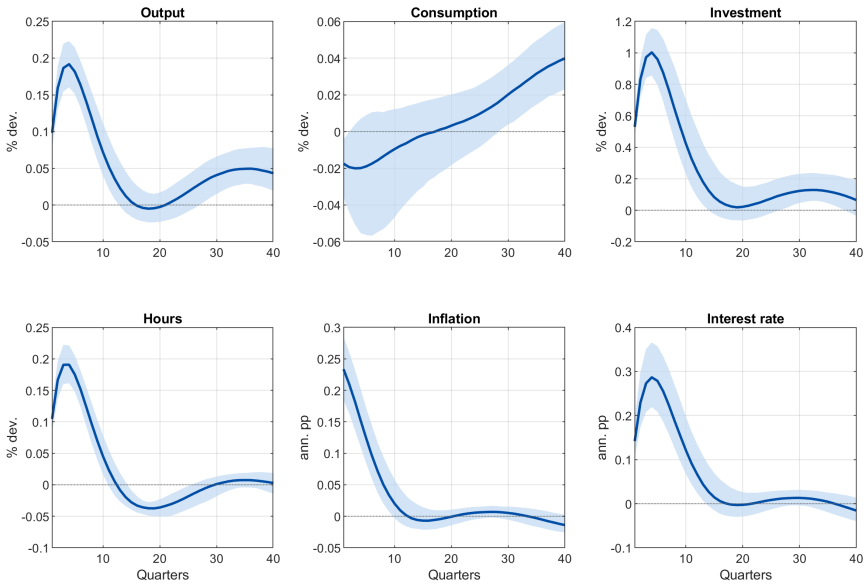


Figure 7: Impulse Responses to an Investment Tax Credit Increase

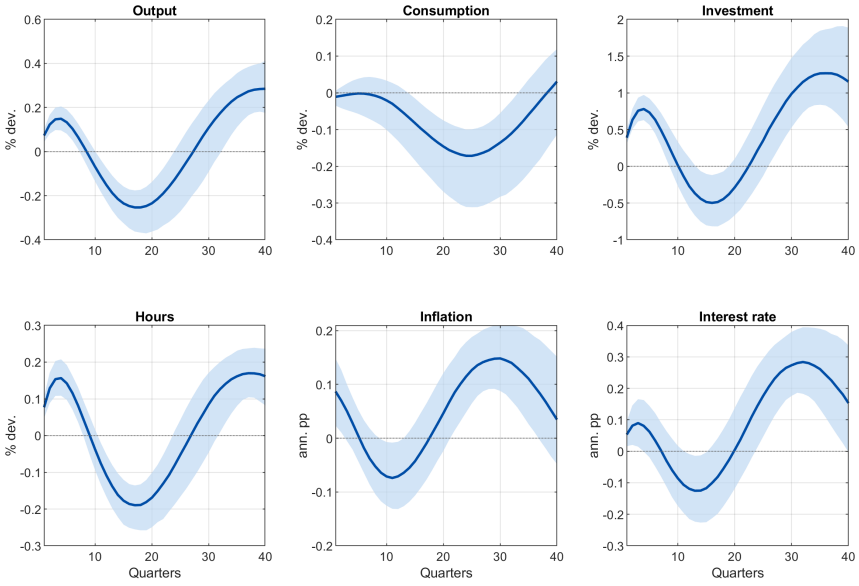


Figure 8: Impulse Responses to an Expensing Rate Increase

Figures for all six instruments: baseline model, one-standard-deviation expansionary shock. Posterior median and 5–95 percentile band. Output, consumption, investment, and hours in percent deviation from steady state; inflation and the interest rate in annualized percentage points.

Appendix

A-1 Alternative Specifications

Table 6 reports the multipliers of the three alternative U.S. specifications with their posterior intervals: AR(1) fiscal rules (Section 2.3), the baseline fiscal rules with the capital-goods friction (Section 2.5), and AR(1) fiscal rules with the friction. Each is estimated separately on the same data and sample as the baseline model, with the same priors for the parameters they share and the same calibration. The AR(1) models have 34 estimated parameters and the models with the baseline rules have 49. All three meet the same convergence criteria as the baseline model.

Table 6: Fiscal Multipliers, Alternative Specifications

Instrument	$H = 1$	$H = 4$	$H = 12$	$H = 20$
Panel A. AR(1) fiscal rules				
Government spending	1.01	0.92	0.70	0.57
	[0.96, 1.04]	[0.83, 0.99]	[0.56, 0.84]	[0.43, 0.74]
Labor tax	0.19	0.41	0.72	0.83
	[0.17, 0.22]	[0.37, 0.44]	[0.68, 0.76]	[0.78, 0.89]
Capital tax	1.26	1.95	2.82	3.02
	[1.16, 1.36]	[1.79, 2.11]	[2.68, 2.96]	[2.92, 3.13]
Consumption tax	0.17	0.26	0.34	0.33
	[0.15, 0.20]	[0.23, 0.29]	[0.31, 0.39]	[0.28, 0.40]
Investment tax credit	0.39	0.87	1.43	1.77
	[0.27, 0.54]	[0.68, 1.09]	[1.19, 1.71]	[1.49, 2.09]
Expensing rate	0.06	0.16	0.32	0.45
	[0.03, 0.09]	[0.13, 0.21]	[0.28, 0.39]	[0.40, 0.55]
Panel B. Baseline fiscal rules with capital-goods friction ($\epsilon_K = 4$)				
Government spending	1.09	1.01	0.50	−0.09
	[1.02, 1.16]	[0.88, 1.17]	[0.29, 0.75]	[−0.38, 0.21]
Labor tax	0.32	0.58	0.55	0.15
	[0.24, 0.40]	[0.45, 0.76]	[0.34, 0.83]	[−0.12, 0.43]
Capital tax	1.39	2.12	2.69	2.65
	[1.26, 1.52]	[1.89, 2.34]	[2.34, 3.12]	[2.21, 3.19]
self-financing	*0%*	*0%*	*7%*	*66%*
Consumption tax	0.22	0.31	−0.01	−0.57
	[0.17, 0.28]	[0.22, 0.42]	[−0.23, 0.23]	[−0.93, −0.25]
Investment tax credit	0.75	1.36	1.68	1.51
	[0.60, 0.91]	[1.11, 1.63]	[1.32, 2.09]	[1.05, 2.03]
self-financing	*0%*	*0%*	*0%*	*28%*
Expensing rate	0.19	0.32	0.07	−0.45
	[0.13, 0.26]	[0.21, 0.46]	[−0.13, 0.30]	[−0.80, −0.18]
Panel C. AR(1) fiscal rules with capital-goods friction ($\epsilon_K = 4$)				
Government spending	1.00	0.93	0.74	0.64
	[0.95, 1.04]	[0.82, 1.00]	[0.60, 0.87]	[0.50, 0.79]
Labor tax	0.21	0.41	0.67	0.76
	[0.18, 0.24]	[0.37, 0.45]	[0.64, 0.71]	[0.73, 0.81]
Capital tax	1.30	1.96	2.67	2.81
	[1.20, 1.42]	[1.78, 2.12]	[2.52, 2.79]	[2.68, 2.91]
Consumption tax	0.16	0.24	0.33	0.34
	[0.13, 0.18]	[0.21, 0.27]	[0.30, 0.37]	[0.30, 0.40]
Investment tax credit	0.49	0.90	1.19	1.36
	[0.36, 0.64]	[0.71, 1.11]	[1.02, 1.42]	[1.18, 1.60]
Expensing rate	0.10	0.20	0.31	0.39
	[0.07, 0.15]	[0.16, 0.27]	[0.26, 0.39]	[0.33, 0.50]

Note: Present-value own-revenue multipliers (equation 3.1). Posterior median with 5–95 percentile intervals in brackets, from 1,000 posterior draws. A positive value means output moves in the expansionary direction of the policy. Rows labeled self-financing give the percentage of posterior draws in which the present value of total tax revenue does not fall at that horizon. It is zero for all other instruments and horizons in Panel B, and for every instrument and horizon in Panels A and C. Each panel is a separately estimated model, 1961Q1–2013Q4. Under AR(1) rules only transfers respond to debt (Section 4).

A-2 United Kingdom

Data

The UK model is estimated on quarterly data for 1997Q2–2016Q4 (79 quarters). Real per-capita consumption, investment, and government consumption are constructed from ONS national accounts (household final consumption, gross fixed capital formation, and general government final consumption), deflated by the GDP deflator and detrended as in the U.S. data. Debt is ONS public sector net debt, inflation is GDP deflator inflation, and the interest rate is Bank Rate. The effective tax rates on labor income, capital income, and consumption follow the same constructions as for the United States Jones (2002); Mendoza et al. (1994), applied to ONS national accounts and public sector finance data; revenue series published without seasonal adjustment are seasonally adjusted before the rates are computed. The model has nine observables: consumption, investment, government consumption, debt, inflation, the interest rate, and the three tax rates.

There is no UK counterpart to the TAXSIM dividend and interest tax series, and UK capital allowances are not directly comparable to the U.S. investment tax credit and bonus depreciation. The dividend tax rate and the expensing rate are therefore not observed, and the UK model has no investment tax credit. The dividend tax and expensing shocks remain in the model and are identified only through cross-equation restrictions.

Model and calibration

The UK model uses AR(1) fiscal rules (equation 2.41) and no capital-goods friction. The steady-state tax rates are UK sample means: $\bar{\tau}^w = 0.215$, $\bar{\tau}^k = 0.522$, and $\bar{\tau}^c = 0.147$. The government consumption share is $\bar{g}/\bar{y} = 0.192$, the World Bank series on general government final consumption over 1997–2016, and the debt ratio is $\bar{b}/\bar{y} = 0.487$, the mean of ONS public sector net debt over the sample. The steady-state expensing, interest income, and dividend tax rates are set at their U.S. values, since no comparable UK series is available; the UK did tax dividends and interest and allowed partial expensing over the sample, so these are approximations. All other calibrated parameters and all priors are as in the U.S. model. With the investment tax credit removed, 32 parameters are estimated. The largest potential scale reduction factor is 1.018 and every parameter meets the convergence criteria used for the U.S. models.

Results

Table 7 reports the UK multipliers and Figure 9 compares them with the U.S. model with AR(1) rules. Section 6 discusses them. The UK expensing multiplier depends on the U.S. steady-state expensing rate and on an unobserved expensing process, and is the least reliable of the UK estimates.

Table 7: Fiscal Multipliers, United Kingdom (AR(1) Fiscal Rules)

Instrument	$H = 1$	$H = 4$	$H = 12$	$H = 20$
Government spending	1.02	0.94	0.68	0.56
	[0.77, 1.18]	[0.70, 1.09]	[0.52, 0.85]	[0.36, 0.78]
Labor tax	0.08	0.25	0.61	0.74
	[0.03, 0.15]	[0.14, 0.39]	[0.46, 0.75]	[0.56, 0.94]
Capital tax	0.92	1.56	2.48	2.61
	[0.73, 1.29]	[1.13, 2.45]	[1.78, 3.44]	[1.87, 3.52]
Consumption tax	0.30	0.38	0.38	0.32
	[0.23, 0.40]	[0.32, 0.46]	[0.30, 0.43]	[0.22, 0.40]
Expensing rate	0.25	0.47	0.73	0.97
	[0.03, 0.39]	[0.22, 0.64]	[0.46, 1.01]	[0.59, 1.40]

Note: Present-value own-revenue multipliers (equation 3.1). Posterior median with 5–95 percentile intervals in brackets, from 1,000 posterior draws. A positive value means output moves in the expansionary direction of the policy. UK model with AR(1) fiscal rules, estimated on 1997Q2–2016Q4 (Appendix A-2). The UK model has no investment tax credit. No posterior draw is self-financing (total tax revenue falls in every draw).

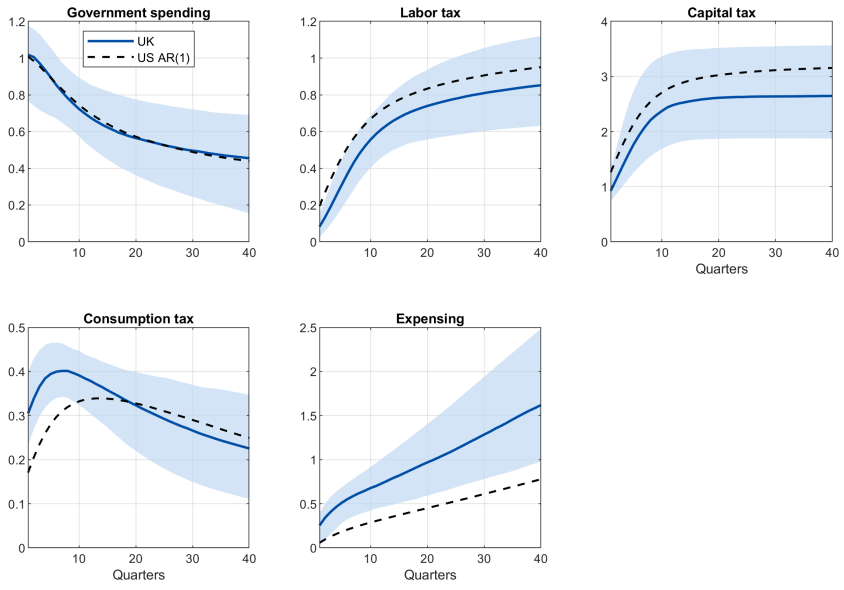


Figure 9: Multipliers by Horizon, United Kingdom and United States (AR(1) Fiscal Rules)
UK: posterior median and 5–95 percentile band. U.S.: posterior median of the model with AR(1) fiscal rules (dashed).

A-3 Household First-Order Conditions

This appendix derives the household first-order conditions of the model presented in Section 2. Zubairy's original derivations, which apply unchanged to the parts of the model that are not modified by our tax extensions, are reproduced in Zubairy (2014). We restrict attention here to the conditions that are either new or modified relative to the baseline.

A-3.1 Utility and Constraints

The representative household maximizes:

$$E_0 \sum_{t=0}^{\infty} \beta^t \, d_t \, U(x_t^c, h_t) \qquad \text{with} \qquad U(x_t^c, h_t) = \frac{1}{1-\gamma} \left[(x_t^c)^a (1-h_t)^{1-a} \right]^{1-\gamma} \tag{A.3.1}$$

where $x_t^c = c_t - b^c s_{t-1}^C$ is habit-adjusted consumption, $s_t^C = \theta^c s_{t-1}^C + (1-\theta^c) c_t$ is the private habit stock, a is pinned down in steady state, γ is risk aversion, and $b^c \in (0,1)$ is the deep-habit parameter.

The household faces the flow budget constraint:

$$\begin{aligned}(1+\tau_t^c) c_t + \frac{b_t}{\pi_t} + (1 - \tau_t^{ic} - \tau_t^k e_{\tau,t}) i_t = (1-\tau_t^w) \frac{w_t}{\tilde{w}_t} h_t + (1-\tau_t^k) r_t^k u_t k_{t-1} + \tau_t^k \delta_\tau k_{t-1}^\tau \\ + \frac{\left[1 + (R_{t-1} - 1)(1 - \tau_t^i)\right] b_{t-1}}{\pi_t} + (1-\tau_t^d) div_t + tr_t\end{aligned} \tag{A.3.2}$$

the capital accumulation law:

$$k_t = (1-\delta) k_{t-1} + i_t \left[1 - \tfrac{\kappa}{2} \left(\tfrac{\mu_t i_t}{i_{t-1}} - 1 \right)^2 \right] \tag{A.3.3}$$

and the tax-basis accumulation law (new):

$$k_t^\tau = (1 - \delta_\tau) k_{t-1}^\tau + i_t (1 - e_{\tau,t}) \tag{A.3.4}$$

The Lagrange multipliers on (A.3.2), (A.3.3), and (A.3.4) are λ_t, $\lambda_t q_t$, and $\lambda_t v_{\tau,t}$ respectively. In Zubairy's baseline the tax basis of capital is not separated from the physical capital stock; here we introduce it because immediate expensing (e_τ) drives a wedge between the two.

A-3.2 Marginal Utility of Consumption

The FOC with respect to c_t gives $\partial U / \partial c_t = (1+\tau_t^c)\lambda_t$. Substituting the marginal utility from (A.3.1):

$$\lambda_t = d_t \, a \, \frac{\left[(x_t^c)^a (1-h_t)^{1-a} \right]^{1-\gamma}}{x_t^c (1+\tau_t^c)} \tag{A.3.5}$$

Relative to Zubairy (2014), the $(1+\tau_t^c)$ factor in the denominator is new. Because λ_t enters every forward-looking equation of the household problem, the consumption tax propagates into the bond Euler, capital Euler, investment FOC, wage Phillips curve, and pricing block without changing the functional form of those equations.

A-3.3 Labor–Leisure Condition

The FOC with respect to h_t combined with (A.3.5) gives the intratemporal condition:

$$d_t(1-a)\frac{\left[(x_t^c)^a(1-h_t)^{1-a}\right]^{1-\gamma}}{1-h_t} = \lambda_t \frac{w_t(1-\tau_t^w)}{\tilde{w}_t} \tag{A.3.6}$$

This is functionally identical to Zubairy's condition after multiplication by $(1+\tau_t^c)$, which reappears through the modified λ_t.

A-3.4 Bond Euler Equation

Nominal bonds pay R_t in period $t+1$ per dollar invested, but only the interest earned, $R_t - 1$, is taxable. Combining the FOC with respect to b_t and applying (A.3.5):

$$\lambda_t = \beta\left[1 + (R_t - 1)(1 - \tau_{i,t+1})\right]\frac{\lambda_{t+1}}{\pi_{t+1}} \tag{A.3.7}$$

Under $\tau_i = 0$ this collapses to Zubairy's Euler equation.

A-3.5 Utilization and Capital Euler

The FOC with respect to utilization u_t delivers:

$$(1-\tau_t^k)r_t^k = \gamma_1 u_t^{\sigma_u} \tag{A.3.8}$$

The FOC with respect to k_t (using the capital constraint) gives:

$$\lambda_t q_t = \beta E_t \lambda_{t+1}\left[(1-\tau_{k,t+1})r_{t+1}^k u_{t+1} - \tfrac{\gamma_1}{1+\sigma_u}(u_{t+1}^{1+\sigma_u} - 1) + q_{t+1}(1-\delta)\right] \tag{A.3.9}$$

The depreciation deduction $\delta q_{t+1}\tau_{t+1}^k u_{t+1}$ present in Zubairy (2014), which credits the capital-tax shield directly in the Euler, is absent here. Instead the shield is carried entirely through the shadow value $v_{\tau,t}$ derived below, avoiding double-counting.

A-3.6 Shadow Value of the Tax Basis

The FOC with respect to k_t^τ delivers a forward-looking recursion. Because an extra unit of tax basis today reduces next-period taxes by $\tau_{k,t+1}\delta_\tau$ and leaves $(1-\delta_\tau)$ units of remaining basis, we obtain:

$$\lambda_t v_{\tau,t} = \beta E_t\left[\lambda_{t+1}\left((1-\delta_\tau)v_{\tau,t+1} + \tau_{k,t+1}\delta_\tau\right)\right] \tag{A.3.10}$$

This equation is new relative to Zubairy (2014) and is what makes it possible to separate immediate expensing (through e_τ) from the ordinary depreciation schedule.

A-3.7 Investment First-Order Condition

The FOC with respect to i_t is the most extensively modified condition. The out-of-pocket cost of a unit of investment falls by τ_t^{ic} (the ITC) and by $\tau_t^k e_{\tau,t}$ (the immediate expensing

wedge). The non-expensed portion $(1-e_{\tau,t})$ generates future depreciation deductions valued at $v_{\tau,t}$:

$$\begin{aligned}\lambda_t(1-\tau_t^{ic}-\tau_t^k e_{\tau,t}) = \lambda_t q_t \left[1-\tfrac{\kappa}{2}\left(\tfrac{\mu_t i_t}{i_{t-1}}-1\right)^2-\kappa\tfrac{\mu_t i_t}{i_{t-1}}\left(\tfrac{\mu_t i_t}{i_{t-1}}-1\right)\right] \\ +\beta E_t\left[\lambda_{t+1}q_{t+1}\kappa\tfrac{\mu_{t+1}i_{t+1}^2}{i_t^2}\left(\tfrac{\mu_{t+1}i_{t+1}}{i_t}-1\right)\right] \\ +\lambda_t v_{\tau,t}(1-e_{\tau,t})\end{aligned} \tag{A.3.11}$$

Under $\tau^{ic}=e_\tau=0$ the LHS and last term collapse to Zubairy's original expression.

A-3.8 Dividend Tax

The dividend tax τ_t^d does not appear in any household first-order condition. Dividends are a residual object taken as given by the household:

$$div_t = y_t - w_t h_t - r_t^k u_t k_{t-1} \tag{A.3.12}$$

The dividend tax only affects the household's after-tax income (equation A.3.2) and enters the government budget constraint through tax revenue.

A-4 Steady-State Derivations

We compute the deterministic steady state analytically by setting all shocks to their unconditional means ($z = d = \mu = 1$; all fiscal innovations equal to zero) and imposing stationarity. Below, a bar denotes a steady-state value and we drop time subscripts.

A-4.1 Shadow Value of the Tax Basis in Steady State

Setting the recursion (A.3.10) to a fixed point and using $\bar{\lambda}_t = \bar{\lambda}_{t+1} = \bar{\lambda}$:

$$\bar{v}_\tau = \frac{\beta \bar{\tau}^k \delta_\tau}{1 - \beta(1 - \delta_\tau)} \tag{A.4.1}$$

which is the present value of the geometric stream of depreciation deductions on one unit of tax basis. This expression is valid for any positive $\bar{\tau}^k$ and δ_τ, and does not depend on the immediate expensing rate.

A-4.2 Tobin's q in Steady State

Setting the investment FOC (A.3.11) to a fixed point (all adjustment cost terms vanish because $\bar{\mu} = 1$ and $\bar{i}_t = \bar{i}_{t-1}$):

$$\bar{q} = 1 - \bar{\tau}^{ic} - \bar{\tau}^k \bar{e}^\tau - \bar{v}_\tau (1 - \bar{e}^\tau) \tag{A.4.2}$$

Under $\bar{\tau}^{ic} = 0$ and $\bar{e}^\tau = 0$, this collapses to $\bar{q} = 1 - \bar{v}_\tau$, matching the geometric-stream expression in Zubairy's baseline.

A-4.3 Real Return on Capital, Real Interest Rate, and Real Wage

The steady-state capital Euler (A.3.9) combined with $\bar{u} = 1$ implies:

$$\bar{r}^k = \frac{\bar{q}}{\beta(1 - \tau^k)} - \frac{q(1 - \delta)}{1 - \tau^k} \tag{A.4.3}$$

The steady-state bond Euler (A.3.7) with $\bar{\pi} = 1$ delivers:

$$\bar{R} = 1 + \frac{1/\beta - 1}{1 - \bar{\tau}^i} \tag{A.4.4}$$

The real wage follows from the firm's optimization conditions and, given the Cobb–Douglas production function, from the capital–labor ratio and marginal cost:

$$\bar{w} = (1 - \theta)\, \overline{mc} \left(\frac{\bar{r}^k}{\theta \overline{mc}} \right)^{-\theta/(1-\theta)} \tag{A.4.5}$$

where $\overline{mc} = (\eta - 1)/\eta$ is pinned down by the desired steady-state markup.

A-4.4 Aggregate Ratios

Given $\bar{h}$, the steady-state capital stock and output follow:

$$\frac{\bar{k}}{\bar{h}} = \left(\frac{\theta \overline{mc}}{\bar{r}^k} \right)^{1/(1-\theta)}, \qquad \bar{y} = \bar{k}^{\theta} \bar{h}^{1-\theta} - \psi \tag{A.4.6}$$

The fixed operating cost ψ is set residually to hit the target dividend share $\overline{div}/\bar{y}$; steady-state investment is $\bar{i} = \delta \bar{k}$; steady-state consumption is $\bar{c} = \bar{y} - \bar{i} - \bar{g}$, using the resource constraint with steady-state utilization cost equal to zero; and steady-state government purchases and debt are determined by the calibrated ratios $\bar{g}/\bar{y}$ and $\bar{b}/\bar{y}$.

A-4.5 Marginal Utility in Steady State

The steady state of the marginal utility equation (A.3.5) pins down the consumption share parameter a jointly with $\bar{\lambda}$. Given $\bar{h}$, $\bar{x}^c = (1 - b^c)\bar{c}$, and the labor–leisure condition (A.3.6), we solve:

$$a = \frac{\bar{x}^c(1+\bar{\tau}^c)}{\bar{x}^c(1+\bar{\tau}^c) + (\bar{w}/\tilde{w})(1-\bar{h})}, \qquad \bar{\lambda} = a \, \frac{\left[(\bar{x}^c)^a (1-\bar{h})^{1-a} \right]^{1-\gamma}}{\bar{x}^c(1+\bar{\tau}^c)} \tag{A.4.7}$$

Both quantities depend on estimated parameters and are recomputed at each draw during Bayesian inference.

A-4.6 Government Budget in Steady State

Setting the government budget constraint (2.31) to a stationary value $\bar{b}$ and using $\bar{\pi} = 1$:

$$\bar{b}\left(1 - \bar{R}\right) + \overline{taxrev} = \bar{g} + \bar{tr} \tag{A.4.8}$$

Given the target debt-to-output ratio and the calibrated tax rates, this determines the steady-state transfer $\bar{tr}$ residually. The debt-residual shock ε_t^b has mean zero and therefore does not appear in the steady state.